Network Estimation Workbook

The theory behind GnssProcessor's network processor, in 41 questions

October 2026

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How the course works

The course teaches the estimation theory of the network processor and its reference choices (datum, S-basis) from this project's own record. It starts on familiar ground, one receiver, and adds one idea at a time. Lessons 1 to 3 work the theory by hand on small examples, then find the same structure in a real run. Lessons 4 to 7 apply it to the project's own choices: the four reference choices, what a user receives, the satellite clock and the integer ambiguities. Lessons 8 and 9 show how the results are tested on simulated and real data and what they support. Every question names the earlier questions it builds on, so the lessons are best worked in order.

Three kinds of question appear in every lesson: DERIVE (a calculation by hand), READ (what a given line of a card, configuration file, log or source file says) and MEASURE (run an existing tool on

an existing run and interpret the number). The last question of the course, SYNTHESIS, gathers what the presentation of 29 October may claim.

1. Work one lesson per sitting, in order.
2. Read the lesson's opening section and figures; open a source only when a question sends there.
3. Write each answer before turning to the answer key. An answer is complete when it names its source.
4. Compare with the key (each question links to its answer and back). Where an answer differs, rerun the key's check.

	Lesson	What it builds	Questions
1	One receiver	the three directions a receiver's data cannot fix	3
2	Two receivers	rank, a reference choice, what it changes and what it cannot, a user, a change of satellites	6
3	The real network	elimination of the stations, two constellations, the count of Eq. 24, the parameter table, the rank gate	5
4	The project's reference choices	the theorem, the four choices and their rows, the code-bias datum, the station level	5
5	What a user sees	what a user absorbs, the header line of the corrections, a declared step, the reference satellite of a double difference	4
6	The satellite clock	the anchor as a step and as a time constant, the hold, two postures, the cost of leaving the final clock	5
7	Ambiguities	why an integer is mostly ionosphere, cycles, integer and float pins, the oracle pin, the corrections at the null	5
8	Simulated data	what a simulation tests, where a run starts, grading in the network's basis, the user's floor	4
9	Real data	the test station and 16 stations, five days, the clock against CODE and GFZ, the claims	4

Before the measure questions. Commands run from `C:/GnssProcessor` in Git Bash, with `C:/Python313/python.exe` (the Python on the path lacks numpy). Several runs and the clock data set are on the D: drive (`D:/GnssProcessorSupport/runs_nom/`, `D:/GnssProcessorSupport/fixtures/`), which must be connected. No question builds the code or starts a new run; all commands together take about two minutes. The status report is `D:/GnssProcessorSupport/presentation/status_update_2026-10-29/status_update_2026-10-29.tex`; paths starting `results/` or `figures/deck/` in captions are in that folder.

Terms. The course uses plain words; the cards, code and configuration files use the record's labels.

In this workbook	In the record	Meaning
our network, the network processor	producer	`runNetwork`: estimates satellite clocks and biases from many stations
corrections, the correction file	the wire, `products.txt`	what the network writes per epoch: C (clock), B (code bias), P (phase bias) and other records
user	consumer	a receiver positioned with the corrections
test station	held-out station	a reference station left out of the network and positioned as a user: WTZZ00DEU (real), SZZ000SIM (simulated)
16 further stations	census	users positioned afterwards from the stored corrections
real data	field	58 IGS stations, 8 May 2024 (day 129)
simulated data set	fixture, sim bed	`runs/simnet60_quiet_col_24h` (quiet), `simnet60_clk_col_24h` (clock error)
reference configuration	golden	`config/golden/golden_field_refound.json` and `golden_sim_refound.json`, generated
case, setting	arm, knob	a copy of a reference configuration with one setting changed; a configuration key
the first hour	ignition	hour 1 after a cold start
reference choice	datum, S-basis, gauge	the constraints that fix the directions no data determine
header line of reference choices	S line	line 2 of `products.txt`
perfect corrections	truth wire	corrections written from the simulator's truth

1 One receiver: three directions the data cannot fix

Goal. Start from the familiar user model and find the first null directions: combinations of unknowns along which every observation stays the same.

The idea. For receiver r, satellite i and signal s on two frequencies, the code and phase equations of the status report (section "One observation model") are

$$P^i_{r,s} = \rho^i_r + c(\mathrm{d}t_r - \mathrm{d}t^i) + T^i_r + \mu_s I^i_r + B_{r,s} - B^i_s + \varepsilon, \tag{1}$$

$$\Phi^i_{r,s} = \rho^i_r + c(\mathrm{d}t_r - \mathrm{d}t^i) + T^i_r - \mu_s I^i_r + \lambda_s N^i_{r,s} + b_{r,s} - b^i_s + \varepsilon, \tag{2}$$

with $\mu_s = f_1^2/f_s^2$ (so $\mu_1 = 1$), I^i_r the slant ionospheric delay on the first frequency, B and b the code and phase biases, and N the ambiguity in cycles. A PPP user takes the orbit and the satellite terms $\mathrm{d}t^i$, B^i_s and b^i_s from a product and estimates the rest. Even then some combinations of the remaining unknowns leave every observation unchanged. Such a combination is a null direction of the design matrix: the data cannot fix a parameter along it, so a convention must. The rest of the course follows these directions from one receiver into a network.

Question 1.1 DERIVE **The receiver clock and the code biases**

answer, p. 23

1. Show that the change $c\,\mathrm{d}t_r \to c\,\mathrm{d}t_r + \delta$, with every receiver bias $B_{r,1}, B_{r,2}, b_{r,1}, b_{r,2}$ changed by $-\delta$, leaves all four rows of a satellite unchanged.
2. Write the two receiver code biases as $B_{r,s} = B_{r,\mathrm{IF}} + \mu_s\, d_r$. Solve for $B_{r,\mathrm{IF}}$ and d_r in terms of $B_{r,1}$ and $B_{r,2}$, and show that $B_{r,\mathrm{IF}}$ is their ionosphere-free combination.

3. Rewrite the code rows with this split. Which combination of clock and code biases do they contain, and what is a user's estimated "receiver clock" therefore?

Question 1.2 DERIVE **The phase bias and the ambiguity**

Builds on 1.1. answer, p. 23

1. Show that, for each signal s, the change $b_{r,s} \to b_{r,s} - \delta_s$ together with $N^i_{r,s} \to N^i_{r,s} + \delta_s/\lambda_s$ for every satellite leaves every phase row unchanged. How many such directions does one receiver have?
2. A PPP user estimates one float ambiguity per satellite and signal and no receiver phase bias. What does that float ambiguity contain, and why is it not an integer even with exact satellite phase biases?
3. What does a difference between two satellites at the same receiver remove, and what must the product supply for that difference to be an integer?

Question 1.3 DERIVE **The ionosphere and the geometry-free code bias**

Builds on 1.1 and 1.2. answer, p. 23

1. With the split of Question 1.1, show that the code rows of every satellite contain the slant ionosphere only as $I^i_r + d_r$.
2. Show that $I^i_r \to I^i_r + \delta$ for every satellite, with $d_r \to d_r - \delta$, leaves the code rows unchanged, and find what absorbs the change on the phase rows.
3. What does a user's estimated slant ionosphere therefore contain? What would have to be fixed by convention for it to be the physical ionosphere?

What this lesson establishes.

- One receiver has three kinds of null direction: the clock against the ionosphere-free code bias; each signal's phase bias against all of that signal's ambiguities; all slant ionospheres against the geometry-free code bias.
- A parameter estimated along such a direction is a combination whose split a convention fixes: here the clock takes $B_{r,\mathrm{IF}}$, the ambiguities take the phase bias, the ionosphere takes d_r.
- A network has the same directions at every receiver, and their mirror images at every satellite (Section 2). For the meeting: these directions are why a network must state its reference choices.

2 Two receivers: rank, reference choice and invariance

Goal. Work the smallest network by hand: find its null directions, remove them with a reference choice, and see what the choice changes and what it cannot change.

The idea. The Mathematical Reference works this example (`docs/report/network_example.tex`, "Worked example: estimability in a two-receiver network"). Receivers A and B observe satellites 1 to 4 (Figure 1). Positions are known; one frequency; no troposphere or ionosphere; no noise. The code collapses to a difference of two clock lumps,

$$P^i_r = a_r - g^i, \qquad a_r = c\,\mathrm{d}t_r + B_r, \qquad g^i = c\,\mathrm{d}t^i + B^i, \tag{3}$$

which is the convention of Question 1.1 already applied: each clock carries its code bias. The phase is

$$\Phi_r^i = a_r - g^i + b^i + N_r^i, \tag{4}$$

with b^i the satellite phase bias and N_r^i the ambiguity in metres, which also absorbs the receiver's phase bias (Question 1.2). The truth, in metres:

$$a_A = 5, \quad a_B = 3; \qquad g = (2, -1, 4, 0.5); \qquad b = (0.10, -0.20, 0.05, 0.15);$$
$$N_A = (10, -5, 20, 8), \qquad N_B = (12, -3, 18, 9).$$

P_r^i	1	2	3	4	Φ_r^i	1	2	3	4
A	3	6	1	4.5	A	13.10	0.80	21.05	12.65
B	1	4	−1	2.5	B	13.10	0.80	17.05	11.65

A third receiver, C, appears in Question 2.5 as a user of the network's corrections.

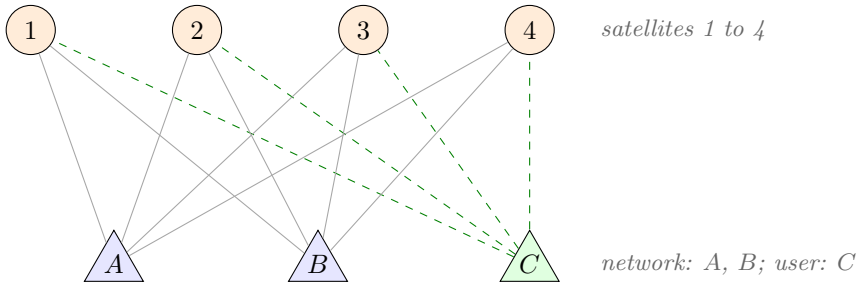


Figure 1: The worked example: two network receivers and four satellites; the user C of Question 2.5 applies the network's corrections.

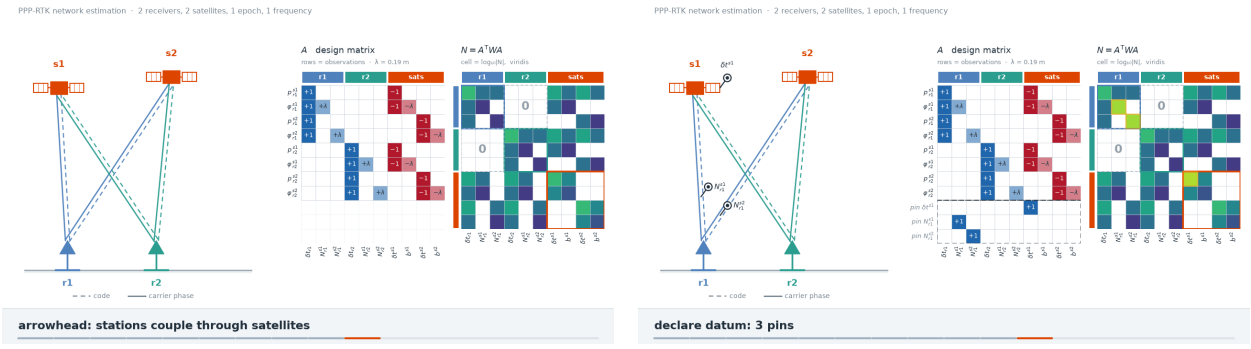


Figure 2: The same network with two satellites, from the project's concept animation (`D:/GnssProcessorSupport/presentation/figures/network_concept.gif`): design matrix $\boldsymbol{A}$ and normal matrix $\boldsymbol{N} = \boldsymbol{A}^{\mathrm{T}}\boldsymbol{W}\boldsymbol{A}$. Left: the empty blocks show that the receivers meet only through satellite terms. Right: eight observations, ten unknowns, three null directions, removed by three pins (the clock of satellite 1 and receiver A's two ambiguities).

Question 2.1 DERIVE The normal matrix and its null vector

Builds on 1.1. answer, p. 24

Order the unknowns $\boldsymbol{x} = [a_A, a_B, g^1, g^2, g^3, g^4]^{\mathrm{T}}$.

1. Write the 8×6 design matrix of the code rows and the normal matrix $\boldsymbol{N} = \boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}$ (unit weights).
2. Show that $\boldsymbol{N}\mathbf{1} = \mathbf{0}$ for the all-ones vector. What is the rank of $\boldsymbol{N}$?
3. What change of the unknowns does $\mathbf{1}$ describe, and why can no number of observations of this kind fix it?

Question 2.2 DERIVE **Two reference choices, one fit**

Builds on 2.1. answer, p. 24

1. Solve the code rows with the pin $a_A \equiv 0$, then with the pin $g^1 \equiv 0$.
2. Subtract the two solutions. What is the difference, in the terms of Question 2.1?
3. Compute $a_A - a_B$, $g^2 - g^1$, $g^3 - g^1$ and $a_B - g^4$ in both solutions. Why do they agree?
4. Under $a_A \equiv 0$, what does the estimated clock of satellite 2 equal in terms of the true clocks?

Question 2.3 DERIVE **Phase: the satellite phase bias needs a second receiver**

Builds on 2.2. answer, p. 25

1. With code and phase together, count the unknowns, the observations, the null directions and the redundancy. Which null directions does the phase add?
2. With receiver A alone, what does each satellite's phase determine, and which pair of unknowns can it not separate? What changes with receiver B?
3. Apply the reference choice $a_A \equiv 0$ with $N_A^i \equiv 0$ for every satellite. Compute the emitted phase bias $\bar{b}^i$ and the estimate $\hat{N}_B^i$ for the four satellites.
4. The physical phase bias of satellite 1 is 0.10 m and the emitted one 10.10 m. What did it absorb? What is $\hat{N}_B^i$?

Question 2.4 DERIVE **Where the hardware biases went**

Builds on 1.1, 1.2 and 2.3. answer, p. 25

Write the network with every hardware bias explicit: receiver and satellite code biases B_r, B^i, receiver and satellite phase biases b_r, b^i, and the integer ambiguities. For f frequencies, n receivers and m satellites, Odijk et al. [2] count six types of null direction (`SBASIS-THEORY-E.md`, section E2): 1a, receiver clock against satellite clock, 1; 1b, receiver against satellite hardware biases, $2f$; 2a, receiver clock against receiver biases, $n-1$; 3a, satellite clock against satellite biases, m; 4, receiver phase bias against ambiguities, $f(n-1)$; 5, satellite phase bias against ambiguities, $f\,m$.

1. Evaluate the six types for $f = 1$, $n = 2$, $m = 4$ and add them.
2. The worked example defined a_r, g^i and N_r^i as lumps. How many unknowns did the lumps remove, and how does the rank change? Which types are gone and which remain? Check the remainder against Question 2.3.

Question 2.5 DERIVE **A user of the emitted corrections**

Builds on 2.2 and 2.3. answer, p. 25

A third receiver C, not in the network, has the truth $a_C = 4$ and $N_C = (11, -4, 19, 10)$ m.

1. Compute its four code and four phase observations.

2. C applies the network's satellite clocks $\hat{g}^i$ (from the choice $a_A \equiv 0$) and phase biases $\bar{b}^i$. Solve for its clock $\hat{a}_C$ and its ambiguities $\tilde{N}^i_C$.
3. What do $\hat{a}_C$ and $\tilde{N}^i_C$ equal in terms of the true values? Whose values does C's solution carry, and does C need to know it?

Question 2.6 DERIVE **When the satellite set changes**

Builds on 2.2. answer, p. 25

1. From the solution with $g^1 \equiv 0$, give the solution under the sum-to-zero choice $\sum_i g^i = 0$ over the four satellites.
2. Satellite 4 sets. Recompute both solutions with satellites 1 to 3. By how much does every clock move under each choice?
3. Instead satellite 1, the pinned one, sets, and the pin moves to satellite 2. By how much does every clock move?
4. Does any of these changes move an estimable quantity? What must a user be told at such an epoch, and why?

What this lesson establishes.

- A reference choice is a set of pins or rows that removes the null directions; every admissible choice fits the data equally well.
- Differences (estimable functions) agree under every choice; the emitted values carry the choice (the satellite clock carries the reference receiver's clock, the phase bias its ambiguity).
- A user applying the corrections inherits the network's reference without knowing it: its estimates are relative to the network.
- A sum-to-zero reference moves whenever a satellite rises or sets; a pin moves only when its own satellite leaves. Either move must be declared.

3 The real network: elimination, counting and the rank gate

Goal. Scale the worked example up to the real network: eliminate the stations, count the null directions, and find them again in a run.

The idea. The network processor solves one joint system per epoch over all stations (status report, step 6). The unknowns shared by all stations, $\boldsymbol{x}_{\mathrm{B}}$, are the satellite clocks, code biases and phase biases; each station r has its own, $\boldsymbol{x}_r$: ambiguities, slant ionosphere, zenith wet delay, receiver clock, inter-system bias and receiver code-bias difference. Stations meet only through $\boldsymbol{x}_{\mathrm{B}}$, so the normal matrix is an arrowhead (Figure 2, left) and each station's block is eliminated exactly before the shared solve:

$$\bar{\boldsymbol{N}}_{\mathrm{BB}} = \sum_r \left(\boldsymbol{N}^{(r)}_{\mathrm{BB}} - \boldsymbol{N}_{\mathrm{B}r} \boldsymbol{N}_{rr}^{-1} \boldsymbol{N}_{\mathrm{B}r}^{\mathrm{T}} \right) + \boldsymbol{N}^{\mathrm{prior}}_{\mathrm{BB}}, \qquad \boldsymbol{x}_r = \boldsymbol{N}_{rr}^{-1} \left(\boldsymbol{u}_r - \boldsymbol{N}_{\mathrm{B}r}^{\mathrm{T}} \boldsymbol{x}_{\mathrm{B}} \right). \tag{5}$$

From epoch to epoch the solution is a Kalman filter in information form. Every epoch a check, the *rank gate*, counts the eigenvalues of $\bar{\boldsymbol{N}}_{\mathrm{BB}}$ below `rank_gate_tol` $\times \lambda_{\max}$ and requires exactly the declared number (`include/network/solve/Sbasis.hpp`, `src/core/network/solve/RankGate.cpp`). A single-receiver PPP filter has the same shape, with epochs in the place of stations (Figure 3).

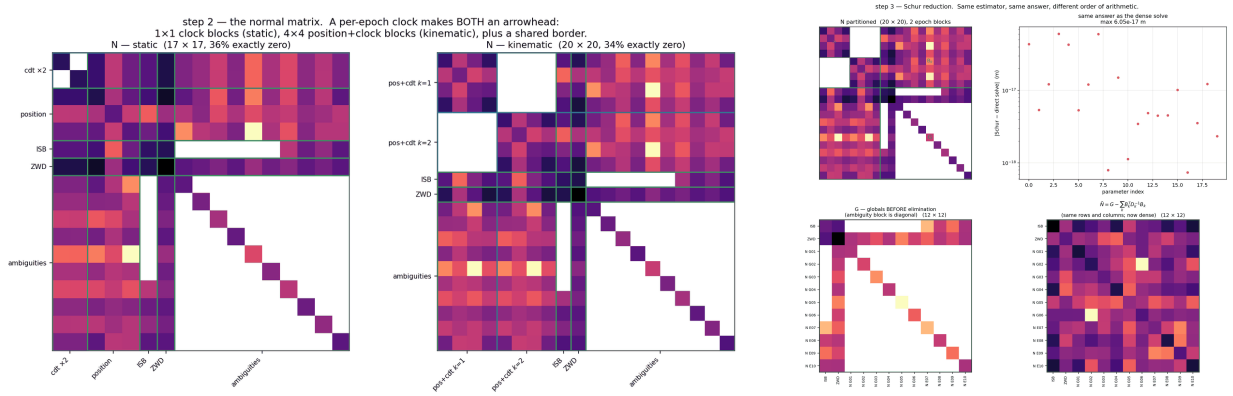


Figure 3: Left: one receiver over two epochs; a clock per epoch makes 1×1 (static) or 4×4 (moving receiver) epoch blocks around a shared border. Right: eliminating the blocks first gives the direct solution to 6×10^{-17} m. September 2026 illustrations.

Question 3.1 DERIVE **Eliminating the receivers**

Builds on 2.1 and 2.2. answer, p. 26

In the code system of Question 2.1, the receiver clocks are the local unknowns and the satellite clocks the shared ones.

1. Compute the reduced shared normal $\bar{\boldsymbol{N}}_{gg} = \boldsymbol{N}_{gg} - \boldsymbol{N}_{ga}\boldsymbol{N}_{aa}^{-1}\boldsymbol{N}_{ag}$ and the reduced right-hand side $\bar{\boldsymbol{u}}_g = \boldsymbol{u}_g - \boldsymbol{N}_{ga}\boldsymbol{N}_{aa}^{-1}\boldsymbol{u}_a$ from the pseudorange table.
2. Find the eigenvalues of $\bar{\boldsymbol{N}}_{gg}$ and its normalized null eigenvector. What weight does it put on each satellite clock?
3. Solve the reduced system with $g^1 \equiv 0$, then back-substitute the receiver clocks with Equation (5). Compare with Question 2.2.

Question 3.2 DERIVE **Two constellations**

Builds on 3.1. answer, p. 26

Add Galileo satellites. Each receiver gets an inter-system bias d_r^E on its Galileo rows ($d_r^G \equiv 0$ by definition: GPS is the reference system), so $P_r^i = a_r + d_r^{g(i)} - g^i$.

1. Find the null directions of this system.
2. After the receiver clocks and inter-system biases are eliminated, how many null directions does the reduced shared normal keep, and what do they look like on the satellite clocks?
3. For 32 GPS and 24 Galileo satellites, what weight does each normalized direction put on each clock?

Question 3.3 DERIVE **Eq. 24 for the real network**

Builds on 1.3 and 2.4. answer, p. 26

1. Sum the six types of Question 2.4 to obtain Eq. 24 of [2], $1 + 2f + (1 + f)(n - 1 + m)$. Add the slant-ionosphere term $n - 1 + m$ (the direction of Question 1.3, now at every receiver and every satellite).
2. Figure 4 uses one constellation of 32 satellites and two frequencies with a slant ionosphere. Check its 369 directions for 60 stations, 361 for 58 and 1,349 for 305.

3. The first epoch of the simulated reference run has $n = 60$, $m = 56$ (32 GPS, 24 Galileo) and $f = 2$. What does Eq. 24 with the slant term give?

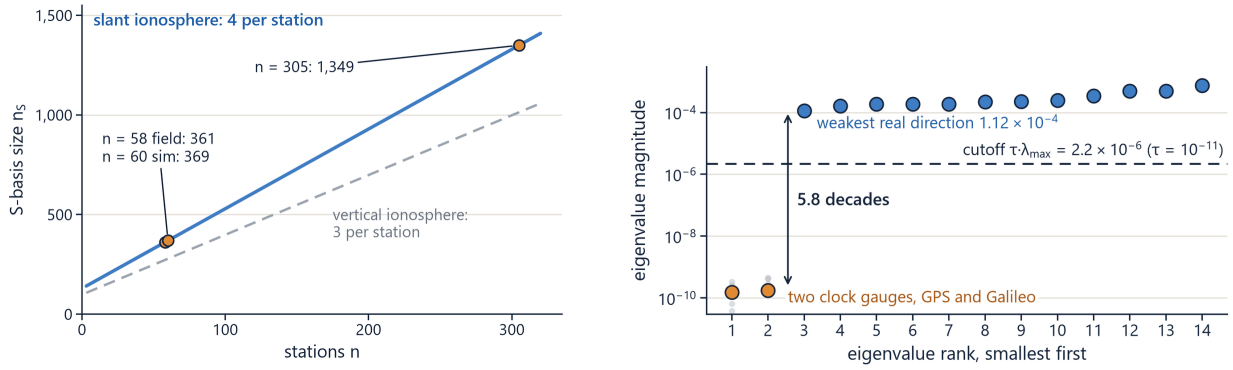


Figure 4: Left: Eq. 24 against the number of stations, one constellation of 32 satellites, two frequencies: about four more directions per station with a slant ionosphere, three with a vertical one (deck slide 6). Right: the rank gate on the first real-data epoch, the 14 smallest eigenvalues of the 414×414 reduced shared normal matrix: two near 10^{-10} (one clock direction per constellation), the weakest real direction 5.8 decades higher, the dashed line the cutoff (deck slide 7).

Question 3.4 READ The parameters as armed

Builds on 1.1 to 1.3 and 3.1. answer, p. 27

Print the table of the simulated reference run (60 stations, GPS and Galileo, two signals each):

```
sed -n '658,686p' runs/_rf0hv15_sim6/stdout.txt
```

1. Map each unknown of Equations (1) and (2) to a row of the table. Which term has no row, and where did Question 1.2 put it?
2. How many shared and how many local columns does the run carry at its first epoch?

Question 3.5 MEASURE The rank gate on the real network

Builds on 3.2 and 3.3. answer, p. 27

Run (about 9 s):

```
C:/Python313/python.exe -I scripts/rf_null_space_read.py runs/_rf0hv15_sim6
grep -n "rank gate evaluated" runs/_rf0hv15_sim6/stdout.txt
```

1. Which two eigenvalues fall below the cutoff at the first epoch, and what weights do their eigenvectors carry? Compare with Question 3.2.
2. How many decades lie between the cutoff and the larger of the two, and between the cutoff and the weakest real direction? Does that meet the placement rule of `src/core/network/solve/RankGate.cpp` (lines 408 to 410)?
3. Eq. 24 gave 465 directions; the gate declares 2. Where did the others go? (Question 2.4 holds the key; Question 4.3 counts the rows.)

What this lesson establishes.

- Eliminating the stations is exact; the reduced shared normal keeps the network's shared null directions.
- With two constellations the shared null space is two flat clock shifts, weight $1/\sqrt{m_g}$ on every clock of system g; the real run shows exactly these, at 0.1767 and 0.2040.
- Eq. 24 counts hundreds of directions; lumps, pins and priors remove almost all of them before the gate, which sees one per constellation. For the meeting: the per-epoch check confirms this count on every epoch.

4 The project's four reference choices

Goal. Name the reference choices the reference configurations make, count their rows, and find the one direction they leave to a weak prior.

The idea. A reference choice is a set of constraints $\boldsymbol{B}^{\mathrm{T}}\boldsymbol{x} = \boldsymbol{0}$, or pins, that picks one solution among those that fit the data equally well. The S-transformation $\boldsymbol{S} = \boldsymbol{I} - \boldsymbol{V}[(\boldsymbol{S}^{\perp})^{\mathrm{T}}\boldsymbol{V}]^{-1}(\boldsymbol{S}^{\perp})^{\mathrm{T}}$ projects onto the estimable space along the null space $\boldsymbol{V}$; its rows are the estimable combinations, and the reference directions carry zero variance (`SBASIS-THEORY-E.md`, section E1, after [2, 3]). The record names four kinds of choice (section E3), each with an analogue in Section 2: a *pin* ($g^1 \equiv 0$); a *prioritized anchor* ($N_A^i \equiv 0$, but elected per satellite and signal); a *sum-to-zero* row ($\sum g^i = 0$); an *ensemble*, an inner constraint over all receivers, measured dead and deleted on 2026-10-14. The reference configurations combine four choices:

Choice	What it fixes	Settings (`golden_sim_refound.json`)	First epoch, simulated
Clock pin	one satellite clock per constellation (the pivot); Questions 2.2 and 3.2	no key (the default); `sticky_pivot`, `handover_stransform`	2 pins (E02, G01)
Prioritized phase anchors, held basis	one ambiguity per satellite and signal, frozen at its basis value; Question 2.3	`phase_datum_prioritized`, `phase_datum_held_basis`, `ambiguity_vehicle float_kf`	112 pins on 18 stations
Code-bias datum	per satellite the ionosphere-free pin and the geometry-free pin at the broadcast T_{GD}; Questions 1.1 and 1.3 at the satellites	`estimate_sat_code_bias`, `sat_gf_datum broadcast_tgd`, `sat_gf_datum_time_linked`	$56 + 56$ rows
Zero-mean receiver DCB prior	the mean of the receivers' geometry-free code biases, σ 10 m; Question 1.3 at the receivers	`rx_code_bias_datum`, `station_prior_before_pd`	120 prior entries on 60 stations

One direction per station and constellation is left to the weak prior: the station's *level*, its slant ionosphere, receiver geometry-free code biases and ambiguities shifted together. It is local, so it is eliminated before the rank gate sees anything.

Question 4.1 READ The theorem in the record

Builds on 2.2. answer, p. 28

Read the boxed theorem of `SBASIS-THEORY-E.md`, section E1, and the paragraph "Datum-invariant even before choosing a datum" of section E2.

1. What exactly is invariant, and under which choices? Where did Question 2.2 show it in six unknowns?
2. Which of these are estimable before any choice: position, zenith delay, vertical ionosphere, a satellite clock, a satellite phase bias?
3. A differential satellite clock wander of about 47 cm was measured before the re-founding. Why could no change of reference cure it, and what kind of change can?

Question 4.2 READ **Four choices in the reference configuration**

Builds on 2.3, 2.6 and 4.1. answer, p. 28

Confirm that the reference configurations are the generated ones, then read `config/golden/golden_sim_refound.json`:

```
C:/Python313/python.exe scripts/make_golden_config.py --verify
```

1. Through which keys does the file arm each choice of the table above?
2. How does each choice behave when a satellite rises or sets, and which of them, by Question 2.6, would move at every rise and set?
3. Which datum-looking keys are absent from the file?

Question 4.3 DERIVE **The rows of the first epoch**

Builds on 3.5 and 4.2. answer, p. 29

The simulated case's first epoch has 56 satellites (32 GPS, 24 Galileo) at 60 stations, two signals per constellation; each satellite carries one clock, two phase biases and two code biases.

1. How many shared columns are there, and how many rows does each choice contribute?
2. Account for the gate's 2: which rows act before the gate, and which after?
3. What does the solver do on an epoch whose count differs, and what would S-basis completion add?

Check against lines 631, 679 to 738 and 4109 to 4112 of `D:/GnssProcessorSupport/runs_nom/_rap_scb_g_q6/stdout.txt`.

Question 4.4 MEASURE **The code-bias datum on the corrections**

Builds on 1.1 and 1.3. answer, p. 29

Read the first epoch's code-bias records of G05 in the simulated reference case and the G05 record of the broadcast navigation file:

```
grep -n -m2 '^B,G05,' D:/GnssProcessorSupport/runs_nom/_rap_scb_g_q6/products.txt
sed -n '562,568p' data/2024-129/products/BRDC00IGS_R_20241290000_01D_MN.rnx
```

1. What is the ratio of the C2W record to the C1C record, what is their ionosphere-free combination, and how does the C1C record compare with cT_{GD}?
2. Where does the satellite's true geometry-free code bias go in the network solution, and what must a user do with its own slant ionosphere for the records to apply?

Question 4.5 DERIVE **The direction the shared check cannot see**

Builds on 1.3, 3.1 and 4.3. answer, p. 30

Card R1D (section 1) gives the loadings of the GPS level direction of station MAJU00MHL: the receiver DCB carries a share 0.0074 of the unit direction, the band-5 code delay 0.2392^2 and the second-channel offset 0.1334^2. The receiver DCB prior has $\sigma = 10$ m.

1. What standard deviation does the prior alone give along the unit level direction? Check the card's "about 116 m".
2. A zero-mean row on the band-5 delay and the second-channel offset at the same σ would add information. By what factor? Check "about eleven times".
3. Card ALD (section 3) gives $v^{\mathrm{T}}Nv = 110$ per unit level for the coupling rows. What σ follows, and which rows hold the level?
4. Why can the rank gate never see this direction, and how did it carry the GPS L5 event into the corrections?

What this lesson establishes.

- No admissible reference choice changes an estimable value or its variance; only estimation (weights, dynamics, model) can.
- The project pins one clock per constellation, anchors one ambiguity per satellite and signal, pins each satellite's code biases to the broadcast group delay, and holds the receivers' code-bias mean with a weak prior.
- The published code biases of the pinned pair carry no estimate; the ionosphere absorbs the true geometry-free bias.
- Each station's level is held by the carried coupling, not by the datum, and the gate cannot see it. For the meeting: the choices are 226 rows and 120 prior entries at the first simulated epoch, not "a few".

5 What a user receives and sees

Goal. Follow the corrections to a user: what it receives, what it cannot see, and what may therefore be graded.

The idea. A user positions itself with the equations of Section 1, with the network's clock C, code bias B and phase bias P in place of the satellite terms, and with its own receiver clock, slant ionosphere and float ambiguities. The three null directions of Section 1, now at the user's receiver, absorb three parts of the corrections: a change common to all clocks of a system, a constant on a satellite's phase bias, and a geometry-free offset of a satellite's code biases. These are the directions a reference choice moves, so a user cannot see the choice (Question 2.5 showed it with numbers). The S-system theory says the same of the correction covariance: its rank-one clock and ionosphere parts "lie in the user's own design range and are absorbed" (`SBASIS-THEORY-E.md`, section E4). Hence standing invariant 5 of `CLAUDE.md`: grade only datum-invariant or single-difference quantities. The correction file declares its choices in a header line (the S line) and marks every change of pivot with an epoch record (`E`, `datum_changed=1`).

Question 5.1 DERIVE **Three things a float user absorbs**

Builds on 1.1 to 1.3 and 2.5. answer, p. 30

Take the user's code and phase rows for satellite i of system g, with receiver clock, inter-system bias, slant ionosphere I^i and float ambiguities as unknowns, and the corrections applied.

1. Show that adding Δ to the clock C of every satellite of system g changes no position.
2. Show that adding a constant d to one satellite's phase bias changes no position, and say what changes instead.
3. Show that adding $(\delta, \mu_2\delta)$ to the two code biases of one satellite is absorbed, and say what that does to the user's ionosphere.
4. The status report's clock comparison (step 5) removes two means before it computes a standard deviation. Which two, and which of the three cases does each stand for?

Question 5.2 READ The header line of reference choices

Builds on 4.2. answer, p. 31

Line 2 of `runs/_rf0hv15_sim6/products.txt` is

```
S,S30000SIM,1,nominal-wu93,0,0,1,0,0.05,1,COD0MGXFIN_20241290000_01D_05M_ORB.SP3+
  COD0MGXFIN_20241290000_01D_30S_CLK.CLK,E:E02,G:G01,cbjump=1,atx=igs20.atx,oref=CoM,
  discfrac=0.25,hsig=pc,cbpin=E:C1C-C5Q,cbpin=G:C1C-C2W,
  sat_gf_datum=broadcast_tgd,pbhold=2
```

(one line in the file). With the writer `src/core/products/NetworkProductIO.cpp` (lines 465 to 602) and `include/products/NetworkProductEmitter.hpp`: what does each field declare? Which fields are the reference choices of Section 4? Which token does a current run add that this one lacks, and which choice does the real-data run's S line leave undeclared?

Question 5.3 MEASURE A declared step, seen and absorbed

Builds on 2.6 and 5.1. answer, p. 32

Find the epochs at which the run declares a change of reference, then measure how every GPS clock record moves across the GPS one (tow 276150):

```
awk -F, '/^E,/ && $7=="1" {print NR": "$0}' runs/_rf0hv15_sim6/products.txt
C:/Python313/python.exe - <<'EOF'
import statistics as st
d, t = {}, None
for line in open('runs/_rf0hv15_sim6/products.txt'):
    f = line.rstrip('\n').split(',')
    if f[0] == 'E':
        t = round(float(f[2]) / 30) * 30
    elif f[0] == 'C' and f[1].startswith('G') and t in (276120, 276150):
        d.setdefault(f[1], {})[t] = float(f[2])
s = [v[276150] - v[276120] for v in d.values() if len(v) == 2]
print(len(s), 'satellites, mean step %.4f m, sd after the mean %.4f m'
      % (st.mean(s), st.pstdev(s)))
EOF
```

1. How large is the step of the GPS clock records, and how much of it is left after the per-system mean is removed?
2. Which move of Question 2.6 is this, and what absorbs it at a user?
3. Why would a grade of raw clock steps call this epoch a 1.6 m defect?

Question 5.4 MEASURE **The reference satellite in every chord**

Builds on 2.2 and 5.1. answer, p. 32

The network's double-difference trace (`network_dd_fractional.csv`) differences every satellite against one reference satellite.

1. Derive: if every satellite's error has variance σ^2 and the reference satellite's σ^2_{ref}, what is the variance of one chord and the correlation between two chords?
2. Run (about 12 s) and compare the L1C error as traced and reference-free (median $|e|$ in cycles, and the share that rounds to the right integer), over all epochs and over the last quarter:

```
C:/Python313/python.exe scripts/simnet_amb_integer.py runs/simnet60_quiet_col_24h \
    --run runs/_rf0hv15_sim6 --ref-free
```

3. Why does a grade on the traced chords report the reference satellite's error?

What this lesson establishes.

- A float user absorbs a system-common clock change, a constant on a phase bias and a geometry-free code-bias offset; the reference choice is invisible to it.
- A declared step of 1.63 m on every GPS clock leaves 3 mm after the per-system mean: grade datum-invariant quantities, never raw steps.
- A double difference built on one reference satellite carries that satellite's error in every chord: grade reference-free. For the meeting: every number in the report is graded this way.

6 The satellite clock in time

Goal. See how the network ties each satellite clock to its past, what the phase-clock hold passes to a user, and why the clock's precision today is the a priori clock product's.

The idea. Three mechanisms shape the per-satellite part of the clock, the part a user cannot absorb.

1. *The temporal anchor* (card Q3, `network.clock_temporal_anchor`). Without it the clock is re-estimated from each epoch's data: the step between epochs was 9.23 cm per 30 s, because the clock's own process noise (`sat_clock_q`, 10^{-3} m^2/s) never binds. The anchor folds the previous epoch's clock in as a prior with variance $\max(\text{var} + q\,\Delta t,\ \text{floor})$ and `clock_anchor_q` $= 2.9 \times 10^{-8}$ m^2/s, a value measured from the clock's structure function (Figure 5, left).
2. *The exact coupling update* (card CTU, `carry_coupling_exact_update`). Carried station rows used to age the clock at 10^{-3} m^2/s while the shared solve folded it at 2.9×10^{-8}; the level froze. The exact update lets it converge at the anchor's rate (Figure 5, right).
3. *The phase-clock hold* (cards PBH and LAH, `emit_pb_hold_phase_clock`). Between declared resets the network keeps each satellite's clock minus phase bias, $C - P$, constant up to a common mode, so a user's phase rows follow the a priori clock.

Card SCB (24 October) made the consequence measurable on a simulated data set with injected satellite clock error (offset σ 0.5 m, rate σ 10^{-5} m/s, random walk 3×10^{-8} m^2/s).

Question 6.1 DERIVE **A spectral density as a step**

answer, p. 32

`clock_anchor_q` is 2.9×10^{-8} m^2/s and the epoch 30 s.

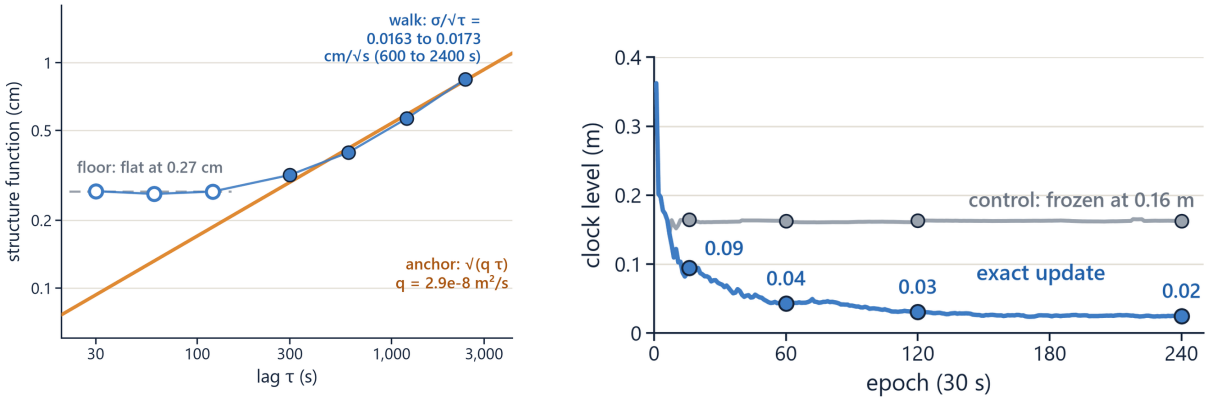


Figure 5: Left: structure function of the network's satellite clock correction against lag: a floor of 0.27 cm from 30 to 120 s, then growth along the anchor's line $\sqrt{q\tau}$, $q = 2.9 \times 10^{-8}$ m^2/s (deck slide 11). Right: simulated network, error of the clock level by epoch, graded from 00:30:15: the control freezes at 0.16 m, the exact coupling update falls to 0.02 m by epoch 240 (deck slide 10).

1. What is the 1σ random-walk step per epoch, and the 1σ accumulation over 6 h?
2. The injected rate on the clock data set has σ 10^{-5} m/s, 0.30 mm per epoch, which card SCB calls "inside the golden anchor's permitted 0.93 mm per epoch". After how many epochs does a 10^{-5} m/s drift exceed the walk's 1σ accumulation? Why does comparing step sizes not guarantee that the anchor follows the drift?

Question 6.2 DERIVE A spectral density as a time constant

Builds on 6.1. answer, p. 32

Treat the anchored clock as a scalar random walk observed once per epoch with noise variance $R = (9.23 \text{ cm})^2/2$ (the pre-anchor step is the difference of two independent epoch estimates). Use the steady-state gain $K = \sqrt{q\,\Delta t/R}$ of `docs/cards/raim-qc/DAY129-FRAME-ANATOMY.md` (lines 138 to 143).

1. What are K and the time constant $\Delta t/K$?
2. What error does a 10^{-5} m/s drift leave? Compare with the coefficient card SCB measured for the reference configuration, 2127 s.
3. What does the same argument give for the self-contained regime's $q = 5 \times 10^{-6}$ m^2/s? Card SCB measured 52 s.

Question 6.3 DERIVE What the phase-clock hold passes to the user

Builds on 5.1 and 6.1. answer, p. 33

`include/network/emit/PbHold.hpp` (lines 60 to 66) writes the held phase bias of a satellite as $P(t) = p(t_0) + [\tilde{c}(t) - \tilde{c}(t_0)]$, with $\tilde{c} = c - M$ the estimated clock minus its system mean.

1. Show what $C - P$ is between resets, and what a user's phase rows see when the a priori clock is $a(t)$ and the truth $a(t) + e(t)$.
2. On the clock data set $e(t)$ is a rate (σ 10^{-5} m/s) plus a random walk (3×10^{-8} m^2/s). What rms error does the hold pass to the phase rows in hour 6 (5 to 6 h after the reset)? Card SCB, section 8, measured 0.201 m for GPS.

3. Lines 311 to 319 of `src/core/config/GnssConfigLoadNetwork.cpp` enforce a precondition on the hold. Which, and why does part 1 make it necessary? Which token of the S line declares the hold in force?

Question 6.4 MEASURE **Two postures on the clock data set**

Builds on 6.2 and 6.3. answer, p. 34

Run the clock grader on the reference case and on the self-contained regime, both on the clock data set (a few seconds each):

```
C:/Python313/python.exe -B scripts/simnet_clock_truth_grade.py \
    --fixture D:/GnssProcessorSupport/fixtures/simnet60_clk_col_24h \
    --run D:/GnssProcessorSupport/runs_nom/_rap_scb_g_k6
C:/Python313/python.exe -B scripts/simnet_clock_truth_grade.py \
    --fixture D:/GnssProcessorSupport/fixtures/simnet60_clk_col_24h \
    --run D:/GnssProcessorSupport/runs_nom/_rap_scb_c_k6
```

Read for each the clock level at hour 6, the rms of the per-satellite mean errors, the correlation of the mean error with the drawn rate, and the user's hourly error in hours 3 to 6. Which posture follows the drift? Why is the reference case's user error so much larger than its clock error? What do these lines not show?

Question 6.5 DERIVE **The cost of leaving the final clock, on real data**

Builds on 6.4. answer, p. 34

Card NBP, section 12 (lines 212 to 219), splits the cost of the native broadcast-clock posture over days 130 to 132 into three steps: the self-contained regime +0.0024 m, removing satellite G01 −0.0007 m, the broadcast clock in place of CODE's +0.0099 m. The pooled median of the whole change is +0.0171 m (line 176).

1. Why do the steps not add to the total? Which part is the clock's information?
2. Against which reference configuration, which days and which hours was the 1.7 cm measured, and what does the first hour add?

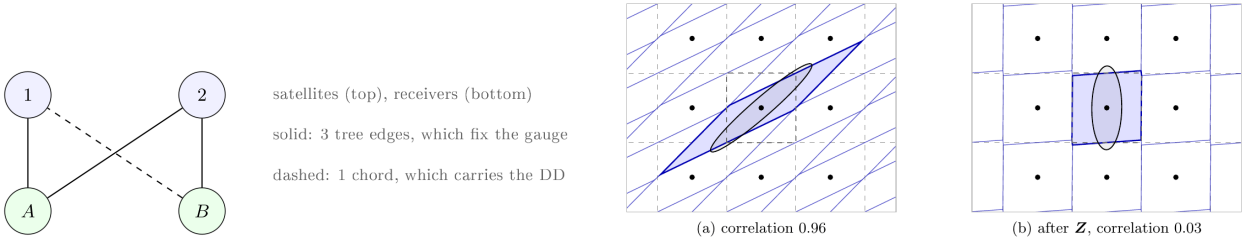
What this lesson establishes.

- $q = 2.9 \times 10^{-8}$ m^2/s is a 0.93 mm step per epoch and, against the epoch noise, a time constant of about 35 min.
- The phase-clock hold hands a user the a priori clock's motion: a wrong a priori clock reaches the phase rows one to one, so the hold is valid only with a precise a priori clock.
- The self-contained postures estimate the clock to about 2 cm on simulated error but are not armed; on real data, replacing CODE's clock costs 1.7 cm at the median. For the meeting: the clock's precision is CODE's.

7 Float ambiguities and integers

Goal. See why the network keeps its ambiguities float, what a complete set of integers would buy, and why the corrections carry no integers yet.

The idea. In the uncombined model every phase row carries both the slant ionosphere and an ambiguity, and on GPS L1 and L2 the two enter in nearly the same direction: a common L1 and L2 cycle moves the phase by $(\lambda_1, \lambda_2) = (0.1903, 0.2442)$ m, the ionosphere along $(1, \mu_2)$ with $\mu_2 = 1.6469$. With a free slant ionosphere per line of sight, the integers become visible only once the double-differenced ionosphere is known to about 0.1 L1 cycle. The re-founding chose the *float vehicle* (card RF-1): each ambiguity is a float state of its station block with a joint covariance, a floored birth prior and zero process noise; integers are graded, not fixed. Question 2.5 showed where a user's integer property comes from: the network's pins.



(a) correlation 0.96 (b) after $\boldsymbol{Z}$, correlation 0.03

Figure 6: Left: the smallest observation graph with a cycle: satellites 1 and 2 (top), receivers A and B (bottom); the three solid arcs form a spanning tree whose ambiguities the reference choice fixes, the dashed chord carries the one double difference the data determine (`docs/report/network_carry.tex`, figure 1). Right: a two-dimensional integer search; at correlation 0.96 the error ellipse is long and the search's pull-in region a long parallelogram, unlike the rounding squares (dashed); after the decorrelating transformation $\boldsymbol{Z}$ the two nearly agree (`docs/report/linear_algebra.tex`, figure 3; a generic example).

Question 7.1 DERIVE **Why an integer is mostly ionosphere**

Builds on 1.2 and 1.3. answer, p. 34

From $f_1 = 1575.42$ MHz and $f_2 = 1227.60$ MHz, compute λ_1, λ_2 and μ_2, the angle between the ionosphere direction $(1, \mu_2)$ and the common-cycle direction (λ_1, λ_2) in metres of phase, its cosine, and the part of one common L1 and L2 cycle that is orthogonal to the ionosphere. Card ARL (line 43) calls a narrow-lane integer "99.3 % ionosphere". Is that a share of amplitude or of variance?

Question 7.2 DERIVE **An ionosphere error in cycles**

Builds on 7.1. answer, p. 34

A double-difference L1 ionosphere error d (metres) is absorbed by the ambiguities.

1. What are the L1 and wide-lane ($L1 - L2$) errors in cycles for $d = 0.4233$ m (no pin) and $d = 0.0419$ m (every ambiguity pinned)? Check the factors against `docs/cards/raim-qc/T3-IONO-MECHANISM.md` (lines 326 to 331).
2. Instantaneous resolution needs about 0.1 L1 cycle. What is that in metres and in TECU, and by what factor does the no-pin case miss it?

Question 7.3 DERIVE **Integer pins and float pins**

Builds on 2.5 and 4.2. answer, p. 35

Work Question 2.5 in cycles, with a receiver phase bias β_r (the same for every satellite), so that $N_r^i = z_r^i + \beta_r$ with z_r^i an integer. Let the network pin $N_A^i \equiv \zeta^i$ instead of 0.

1. What does the network emit as $\bar{b}^i$, and what does the user C estimate as $\tilde{N}_C^i$?

2. Form the difference between satellites i and k at C. When is it an integer?
3. The reference configuration pins each anchor ambiguity at its own float value (`docs/network_ambiguity_resolution_design.md`, line 335). What does part 2 predict for its corrections? What does card IRD (sections 1 to 3) propose instead?

Question 7.4 READ **Complete and partial pins**

Builds on 7.2. answer, p. 35

Read sections 0 and 8.1 of `docs/SPEC_oracle_pin.md`. What is the "integer-information deficit", which row of its table carries it, and why is the raw per-line-of-sight error (2.73 m) the wrong row? What do the partial pins (L5Q alone; L1C with L2W; one station; one satellite) show, and on which simulated data set were the numbers produced?

Question 7.5 MEASURE **Integers in the float, none in the corrections**

Builds on 7.3. answer, p. 35

Run (each under 10 s):

```
C:/Python313/python.exe scripts/simnet_wire_integer_consistency.py \
    runs/simnet60_quiet_col_24h --run runs/_rf0hv15_sim6 --bands L1C,L2W \
    --systems G --wl --nl --null-mc 500 --start-hms 00:30:15
C:/Python313/python.exe scripts/simnet_producer_ar_grade.py runs/simarl_r2 \
    --run runs/_rap_arl_r2dp --settle-h 1
```

The first grades the published GPS phase biases for integer consistency (the fraction of wide-lane and narrow-lane combinations against a uniform null; a set of perfect corrections reads 0.000). The second grades the log-only integer search of card ARL. What does each print, and what do the two together say about where the integers are?

What this lesson establishes.

- An L1 and L2 integer step lies 6.66° from the ionosphere direction; with a free slant ionosphere the integers need a double-differenced ionosphere to 0.117 TECU.
- A user's between-satellite ambiguity is an integer only if the network's pins are integers; the reference configuration pins float values, so its corrections sit at the uniform null.
- The network's float solution holds integers (a log-only search fixes 42 % on a simplified simulation, none wrongly). For the meeting: no integers are published or fixed on real data.

8 Testing on simulated data

Goal. See what the simulated network certifies, where a simulated run starts and is graded, and how much error the network adds above the user's own floor.

The idea. The simulated data set `runs/simnet60_quiet_col_24h` holds 60 network stations and the test station SZZ000SIM at 30 s from 00:00:30 (Figure 7, left). It is *quiet*: no cycle slips and no satellite clock error, because its truth clock is the CODE product that the network also takes as its a priori clock. The simulator extrapolated the satellite positions outside its SP3 window for its first epochs; the reference case is graded from 00:30:15. The network reports clocks and biases in its own reference choices, so truth is moved into that basis before any comparison. The simulator also writes a set of perfect corrections, so the same test station can be positioned with both sets (Figure 7, right).

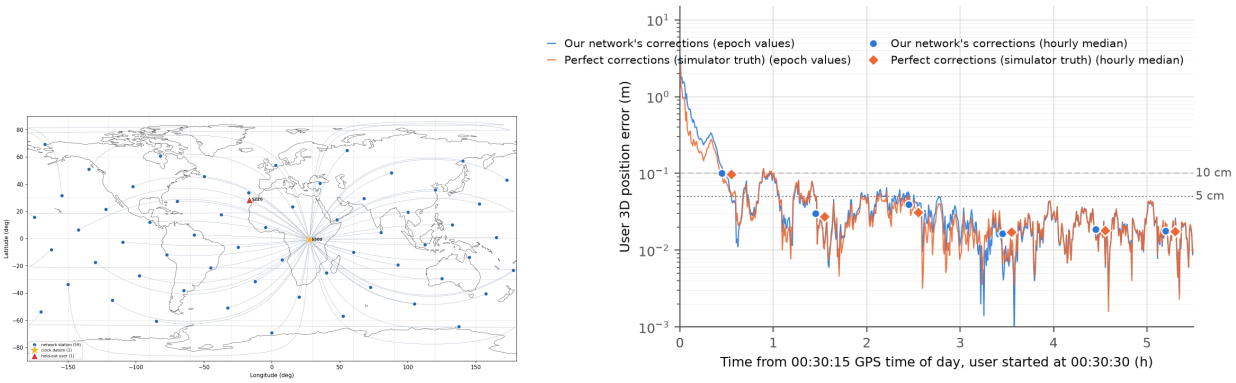


Figure 7: Left: the simulated network, 59 stations and the clock-datum station S30000SIM (star); the test station SZZ000SIM (triangle) is left out (deck slide 15). Right: SZZ000SIM with our network's corrections (blue) and with perfect corrections (orange), user started at 00:30:30; lines are epochs, markers hourly medians (`results/t2/plain/t2_sim_floor_restart.png`).

Question 8.1 READ What the quiet simulation cannot test

Builds on 6.3. answer, p. 36

Read card SCB, section 1 and the start of section 2 (lines 17 to 25), and `results/t2/results.md` (lines 123 to 126). What makes the clock untestable on the quiet data set, and what does its 0.9 cm agreement certify instead? Check the absence of injected clock error directly:

```
ls runs/simnet60_quiet_col_24h/*/truth_satclock.csv 2>/dev/null | wc -l
ls D:/GnssProcessorSupport/fixtures/simnet60_clk_col_24h/*/truth_satclock.csv | wc -l
```

Question 8.2 DERIVE Where a simulated run starts

answer, p. 36

1. The data set's first epoch is tow 259230 (00:00:30). An epoch is reached when its elapsed time since the first epoch is at least the offset (`include/math/TimeUtils.hpp`, lines 416 to 448). At which tow and time do `network.start_offset_hours` 0.504167 and 0.4958333333 start a run? What does the "half-epoch back-off" do?
2. Count the extrapolated epochs. The geometry log warns only when the extrapolation exceeds half the SP3 interval (`src/core/model/ModelGeometry.cpp`, line 1204):

   ```
   grep "Position extrapolation" \
       runs/simnet60_quiet_col_24h/SZZ000SIM/logs/geometry.log \
     | sed -E 's/.*tow ([0-9.]+):.*/\1/' \
     | awk '{c[int(($1+15)/30)*30]++} END{for(e in c) print e,c[e]}' | sort -n
   ```

   How many epochs are logged, how many precede the SP3 window (which opens at tow 259500), and is "faulty in the first 30 minutes" the right description?

Question 8.3 MEASURE The published baseline's clock in the network's basis

Builds on 4.4, 5.1 and 8.2. answer, p. 36

Run the clock grader on the published simulated baseline (started 00:00:30) and on card SCB's quiet case (started 00:31:00):

```
C:/Python313/python.exe -B scripts/simnet_clock_truth_grade.py \
    --fixture runs/simnet60_quiet_col_24h --run runs/_rf0hv15_sim6 --hours 6
C:/Python313/python.exe -B scripts/simnet_clock_truth_grade.py \
    --fixture runs/simnet60_quiet_col_24h \
    --run D:/GnssProcessorSupport/runs_nom/_rap_scb_g_q6
```

Compare the clock level by hour and the rms of the per-satellite mean errors. Rerun the first with `--no-ds`: what does the datum share remove, and why is it added to the truth rather than subtracted from the estimate? Why does the first run's user not show its per-satellite constants?

Question 8.4 DERIVE **The network's cost and the user's floor**

Builds on 8.2. answer, p. 37

From `results/t2/results.md` (lines 195 to 206), the hourly 3D medians with the user restarted at 00:30:30, in metres: perfect corrections 0.0983, 0.0275, 0.0311, 0.0173, 0.0181, 0.0176; our network 0.1013, 0.0301, 0.0396, 0.0166, 0.0186, 0.0178. Compute the network's cost per hour, the largest cost and the floor in hours 4 to 6. In which component does the floor sit (`results/t2/metrics.csv`)?

What this lesson establishes.

- The quiet simulation has no clock error to estimate; it certifies the algebra and the biases.
- Nine epochs are extrapolated; 00:30:15 is a grading origin, and a run started at 00:00:30 keeps per-satellite clock constants of 0.32 m that no float user can see.
- With perfect corrections as the control, the network adds at most 0.9 cm; the 1.7 to 1.8 cm floor is the user side's, in the height. For the meeting: one realization, no slips, no clock error.

9 Real data, and what may be claimed

Goal. Grade the real-data result the way the project does, and put the course together into what the presentation may claim.

The idea. The real-data reference case (`config/golden/golden_field_refound.json`, run `runs/_rf0hv15_field6`) processes 58 IGS stations on 8 May 2024, GPS L1/L2 and Galileo E1/E5a. The test station WTZZ00DEU is left out of the network and positioned as a moving receiver from the clock and bias records, without integer fixing and with its own float ionosphere; it shares its site with two network stations. Sixteen further stations, none of them in the network, are positioned from the same file afterwards. A change is graded against the baseline pair and an unchanged copy run by the same software, on the test station by hour and on the 16 stations, never on one alone; the first hour is excluded from settled claims (`CLAUDE.md`, "Network runs: start from a golden").

Question 9.1 MEASURE **The test station and the 16 stations**

Builds on 5.4. answer, p. 37

Run (about 2 s and 14 s):

```
C:/Python313/python.exe scripts/rapid_arm.py score \
    runs/_rf0hv15_field6 runs/_rf0hv15_sim6
C:/Python313/python.exe scripts/clock_level/gcensus.py grade galpc galp galpg
```

Read the test station's median error by hour on both data sets, the 16 stations' median by hour and the paired Galileo result. By what factor is hour 1 worse than the mean of hours 2

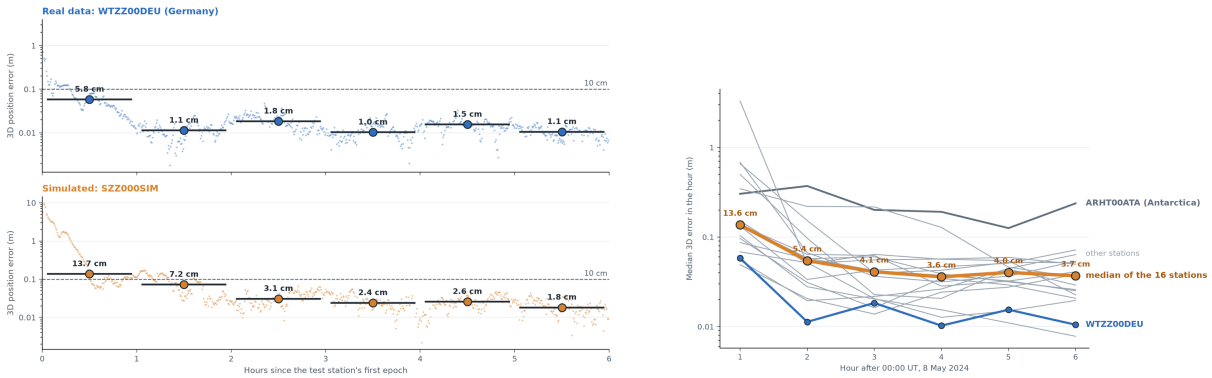


Figure 8: Left: the test station with our network's corrections, real data (WTZZ00DEU, top) and simulated (SZZ000SIM, bottom); dots are epochs, bars hourly medians. Right: hourly median for the 16 further stations (grey), their median (orange), WTZZ00DEU (blue) and ARHT00ATA (dark). Status report figures.

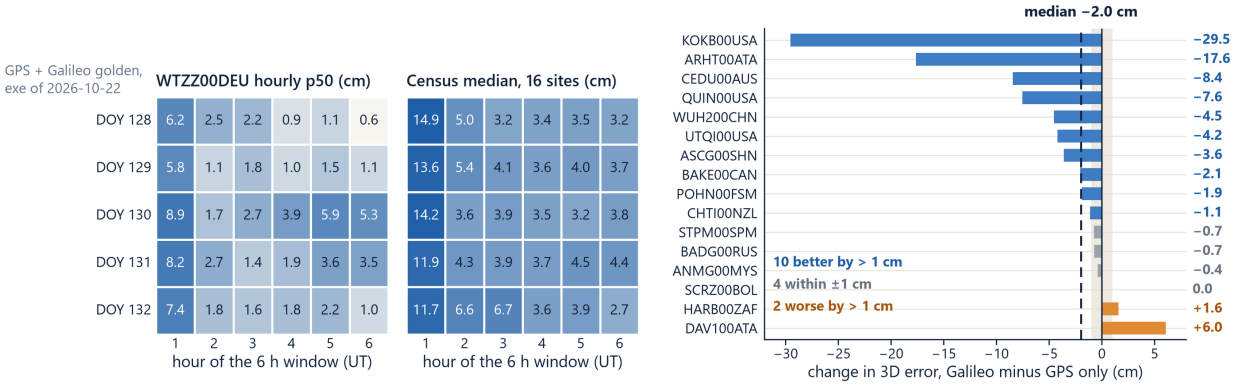


Figure 9: Left: five days, 7 to 11 May 2024, hourly median in cm: WTZZ00DEU and the median of the 16 stations (the labels "golden" and "census" are the record's names for the reference configuration and the 16 stations; deck slide 33). Right: change of each station's error in hours 2 to 6 when Galileo is added, median −2.0 cm (deck slide 19).

to 6, at WTZZ00DEU and at the median station? In which mode is WTZZ00DEU positioned (`runs/_rf0hv15_field6/arm.json`: `filter.process_noise.position_q`, `network.closed_loop.ar_mode`), and which network stations share its site?

Question 9.2 DERIVE Five days from the stored grades

Builds on 9.1. answer, p. 37

`D:/GnssProcessorSupport/runs_nom/mday_galileo_grades.txt` holds, per day 128 to 132, the 16 stations' median by hour and WTZZ00DEU by hour. From hours 2 to 6 of the five days, compute the median and the mean of the 25 sixteen-station medians and the median of the 25 WTZZ00DEU values. On which day is WTZZ00DEU worse than the median station in hours 4 to 6, and what does that do to "the best station"?

Question 9.3 READ The clock against CODE and GFZ

Builds on 5.1 and 6.3. answer, p. 38

Read `results/t3/table_clock_report.tex` and `results/t3/results.md` (the summary table

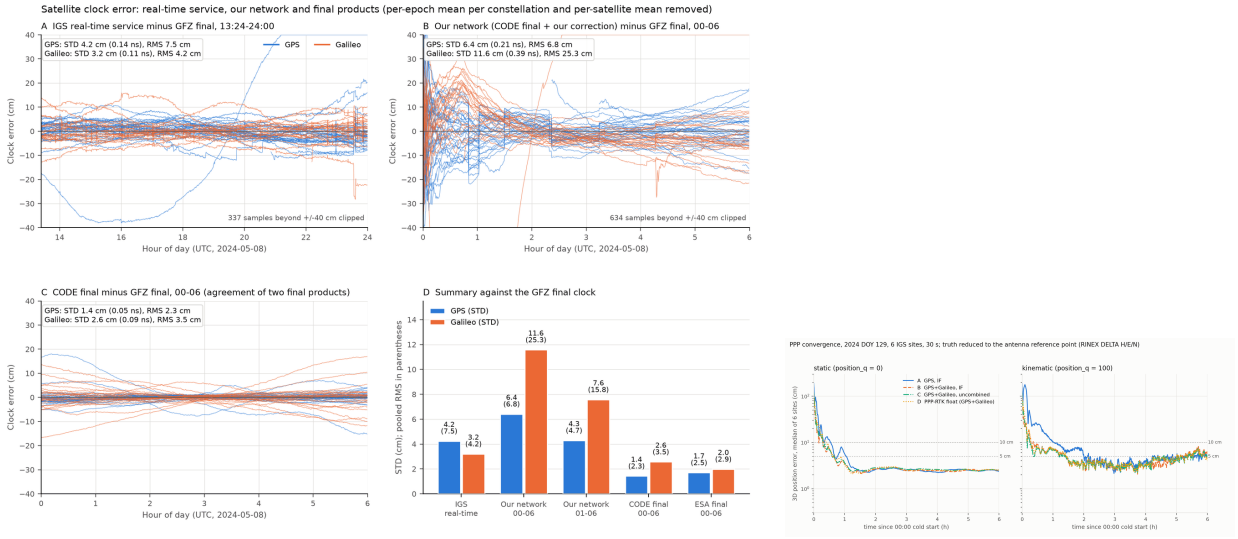


Figure 10: Left: satellite clock error against the GFZ final clock after removing the per-system epoch mean and each satellite's mean: (A) IGS real-time service, (B) our network, CODE final plus our correction, 00:00 to 06:00 with the first hour, (C) CODE final, (D) summary (`results/t3/fig_clock_limits_plain.png`). Right: the PPP comparison of steps 1 to 3, median over six stations, static and moving receiver (`results/t1/fig_t1_convergence.png`).

and lines 100 to 113). Which clock is compared with which, over which hours? Why is GFZ the reference and not CODE? Over the same hours, how does the network's corrected clock compare with CODE's alone? What do the correlations of the correction with (GFZ minus CODE) and (ESA minus CODE) say?

Question 9.4 SYNTHESIS **What the presentation may claim**

Builds on the whole course. answer, p. 38

For each of five topics, state in two or three sentences what the presentation of 29 October may claim and the evidence that bounds it: (a) the model and its per-epoch check; (b) the reference choices; (c) the satellite clock; (d) integer ambiguities; (e) the simulated and real-data results.

What this lesson establishes.

- On 8 May 2024 the test station reaches 1.0 to 1.8 cm and the median of the 16 further stations 3.6 to 5.4 cm in hours 2 to 6, without integers.
- Over five days the median station holds 3.8 cm; the test station is the best station on some days, not all, and shares its site with network stations.
- The network's corrected clock is noisier than CODE's alone and carries no clock information: the user's precision is CODE's.

Answers

Each answer gives its source: the card and lines, the file and lines, or the run. Where a command reproduces the answer, *Check* gives it. Answers marked *checked* were reproduced on 24 October 2026 (the list is in the last section). Cards named by a short name are files in `docs/refound/`: ALD `ALD_anchor_level_datum_2026-10-23.md`, ARL `ARL_integer_ambiguity_ladder_2026-10-09.md`, CTU `CTU_coupling_time_update_2026-09-28.md`, FCFC `FCFC_fixed_cycle_ledger_2026-10-11.md`, GAD `GAD_galileo_adoption_2026-10-13.md`, IRD `IRD_integer_recovering_datum_2026-10-13.md`, LAH `LAH_level_aware_hold_2026-09-29.md`, NBP `NBP_native_broadcast_clock_posture_2026-10-07.md`, R1D `R1D_station_level_datum_2026-10-23.md`, SCB `SCB_sim_clock_error_bed_2026-10-24.md`, SCL `SCL_self_contained_clock_level_2026-10-03.md`. The S-system cards `SBASIS-THEORY-E.md` and `SBASIS-CONTRACT.md` are in `docs/cards/network-vehicle/`. "The worked example" is `docs/report/network_example.tex` (Mathematical Reference, "Worked example: estimability in a two-receiver network"). Status report sections are cited by name; its wording was qualified on 24 October with the boundaries of Question 9.4.

Lesson 1: one receiver

Answer 1.1 (question, p. 3)

1. Every code row contains $c\,\mathrm{d}t_r + B_{r,s}$ and every phase row $c\,\mathrm{d}t_r + b_{r,s}$; each sum gains $\delta - \delta = 0$. One receiver has one such direction.
2. $B_{r,1} = B_{r,\mathrm{IF}} + d_r$ and $B_{r,2} = B_{r,\mathrm{IF}} + \mu_2 d_r$ give $d_r = (B_{r,2} - B_{r,1})/(\mu_2 - 1)$ and $B_{r,\mathrm{IF}} = (\mu_2 B_{r,1} - B_{r,2})/(\mu_2 - 1) = (f_1^2 B_{r,1} - f_2^2 B_{r,2})/(f_1^2 - f_2^2)$, the ionosphere-free combination. d_r is a scaled receiver DCB.
3. $P^i_{r,s} = \cdots + (c\,\mathrm{d}t_r + B_{r,\mathrm{IF}}) + \mu_s(I^i_r + d_r) + \ldots$: the clock appears only as $c\,\mathrm{d}t_r + B_{r,\mathrm{IF}}$. A user's "receiver clock" is this ionosphere-free clock. The network defines the satellite clock the same way: "the ionosphere-free clock (clock + IF code bias)".

Source. status report, "One observation model"; `SBASIS-THEORY-E.md` lines 87 to 92 (the equations) and 158 to 160 (the ionosphere-free clock).

Answer 1.2 (question, p. 4)

1. $\lambda_s(N + \delta_s/\lambda_s) + (b_{r,s} - \delta_s) = \lambda_s N + b_{r,s}$ for every satellite. One direction per signal: two for a dual-frequency receiver (type 4 of the network table, which counts $f(n-1)$ of them).
2. It estimates $N^i_{r,s} + b_{r,s}/\lambda_s$ (and, without satellite phase biases, $-\,b^i_s/\lambda_s$ as well). The receiver phase bias is a fraction of a cycle common to all satellites, so the float ambiguity is not an integer however exact the satellite biases are. The same absorption makes "a constant satellite phase bias simply join the ambiguity, so case D matches case C" (status report, step 3).
3. The difference between two satellites removes $b_{r,s}$, common to both. It is an integer only if both satellites' phase biases are removed exactly: the product must publish them (the observable-specific biases of step 3).

Source. `SBASIS-THEORY-E.md` lines 97 to 104 (type 4); status report, step 3.

Answer 1.3 (question, p. 4)

1. $\mu_s I_r^i + B_{r,s} = B_{r,\mathrm{IF}} + \mu_s(I_r^i + d_r)$.
2. On code, $\mu_s(I + \delta + d - \delta)$ is unchanged. On phase the shifted ionosphere adds $-\mu_s\delta$ to every satellite's row; the float ambiguities absorb it, $N_{r,s}^i \to N_{r,s}^i + \mu_s\delta/\lambda_s$ for every satellite. One more direction per receiver.
3. The estimated slant ionosphere is $I_r^i + d_r$ (and, with satellite terms from a product, minus that product's satellite geometry-free bias). It is the physical ionosphere only if d_r is fixed by a convention, such as a zero-mean condition on the receivers' DCBs, or separated by a vertical ionosphere model. This is Odijk's $\tilde{\iota} = \iota + d_{r,\mathrm{GF}} - d_{\mathrm{GF}}^s$ (types 2b and 3b), and the reason for the network's zero-mean receiver DCB prior (Section 4).

Source. `SBASIS-THEORY-E.md` lines 122 to 128.

Lesson 2: two receivers

Answer 2.1 (question, p. 5)

1. Each code row has $+1$ in its receiver's column and -1 in its satellite's. $\boldsymbol{N} = \boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}$ has 4 on the receiver diagonal (each receiver clock is in four rows), 2 on the satellite diagonal, -1 between every receiver and every satellite clock, and 0 elsewhere:
$$\boldsymbol{N} = \begin{bmatrix} 4\boldsymbol{I}_2 & -\boldsymbol{J}_{2\times 4} \\ -\boldsymbol{J}_{4\times 2} & 2\boldsymbol{I}_4 \end{bmatrix}, \qquad \boldsymbol{J} \text{ all ones.}$$
2. Every row sums to zero ($4 - 4 = 0$; $-1 - 1 + 2 = 0$), so $\boldsymbol{N}\mathbf{1} = \mathbf{0}$; the rank is 5.
3. $\mathbf{1}$ adds the same κ to every clock, receiver and satellite. Each observation is a difference $a_r - g^i$, so none changes, and no number of such observations fixes the common level: one reference choice is needed (type 1a). *Checked.*

Source. the worked example, eq. ex-code-normals.

Check.

```
C:/Python313/python.exe - <<'EOF'
import numpy as np
A = np.array([[1 if j == r else 0 for j in range(2)]
              + [-1 if k == i else 0 for k in range(4)]
              for r in range(2) for i in range(4)])
N = A.T @ A
print(N, np.linalg.matrix_rank(N), N @ np.ones(6))
EOF
```

Answer 2.2 (question, p. 6)

1. $a_A \equiv 0$: $g^i = -P_A^i$, $a_B = a_A - (P_A^1 - P_B^1)$: $a_B = -2$, $g = (-3, -6, -1, -4.5)$. $g^1 \equiv 0$: $a_A = P_A^1 = 3$, $a_B = P_B^1 = 1$, $g^i = a_A - P_A^i = (0, -3, 2, -1.5)$.
2. The second minus the first is 3 on every unknown: $3 \cdot \mathbf{1}$, a vector of the null space.
3. $a_A - a_B = 2$, $g^2 - g^1 = -3$, $g^3 - g^1 = 2$, $a_B - g^4 = 2.5$ in both, and in the truth. Each combination $\boldsymbol{f}$ has $\boldsymbol{f}^{\mathrm{T}}\mathbf{1} = 0$, so adding a multiple of $\mathbf{1}$ cannot change it: the datum-invariance theorem in six unknowns (Question 4.1).

4. $\hat{g}^2 = g^2 - a_A = -1 - 5 = -6$: the true clock shifted by the reference receiver's clock. This lump is why a network declares its choice. *Checked.*

Source. the worked example, tables "Two datum choices, same fit" and "What is invariant".

Answer 2.3 (question, p. 6)

1. Unknowns $6 + 4 + 8 = 18$; observations 16; null directions 5: the clock direction of Question 2.1 and one per satellite, $(b^i + c_i,\ N_A^i - c_i,\ N_B^i - c_i)$; rank 13; redundancy $16 - 13 = 3$, the three code double differences $((n-1)(m-1))$, identically zero in this noise-free model.
2. With A alone each satellite's phase gives one equation in $b^i + N_A^i$: two unknowns that cannot be separated. With B there are two equations in three unknowns, rank 2; the new estimable quantity is $N_A^i - N_B^i$, the between-receiver single difference.
3. With $a_A \equiv 0$, $a_A - g^i = P_A^i$, so $\bar{b}^i = \Phi_A^i - P_A^i = (10.10, -5.20, 20.05, 8.15)$ and $\hat{N}_B^i = \Phi_B^i - P_B^i - \bar{b}^i = (2, 2, -2, 1)$.
4. $\bar{b}^1 = b^1 + N_A^1 = 0.10 + 10$: the emitted bias carries the pinned receiver's ambiguity, Odijk's $\bar{b}^i = b^i +$ (pivot-receiver ambiguity). $\hat{N}_B^i = N_B^i - N_A^i$, the between-receiver single difference an RTK user fixes. *Checked.*

Source. the worked example, "Adding carrier phase".

Answer 2.4 (question, p. 6)

1. $1 + 2 + 1 + 4 + 1 + 4 = 13$, which is Eq. 24 with $f = 1$, $n = 2$, $m = 4$: $1 + 2 + 2(1 + 4) = 13$.
2. The full model has 26 unknowns (clocks 6, receiver and satellite code biases $2 + 4$, receiver phase biases 2, satellite phase biases 4, ambiguities 8); the lumped model 18. The lumps remove 8 unknowns (B_r, B^i, b_r) without changing the column space, so the rank stays 13 and the null space drops from $26 - 13 = 13$ to $18 - 13 = 5$. Gone: types 1b (2), 2a (1), 3a (4) and 4 (1), eight in all. Left: 1a (1) and 5 (4), the five of Question 2.3. A lump is itself a reference choice: defining $a_r = c\,\mathrm{d}t_r + B_r$ fixes the split along the removed directions. *Checked.*

Source. `SBASIS-THEORY-E.md` lines 97 to 114 (the table and Eq. 24).

Answer 2.5 (question, p. 6)

1. $P_C^i = a_C - g^i = (2, 5, 0, 3.5)$; $\Phi_C^i = a_C - g^i + b^i + N_C^i = (13.10, 0.80, 19.05, 13.65)$.
2. Code: $\hat{a}_C = P_C^i + \hat{g}^i = -1$ for every satellite. Phase: $\tilde{N}_C^i = \Phi_C^i - \hat{a}_C + \hat{g}^i - \bar{b}^i = (1, 1, -1, 2)$.
3. $\hat{a}_C = a_C - a_A = 4 - 5$, through $\hat{g}^i = g^i - a_A$; $\tilde{N}_C^i = N_C^i - N_A^i$, through $\bar{b}^i = b^i + N_A^i$. C's solution carries receiver A's clock and ambiguities, and C needs no knowledge of A: "the user never needs to know the network pivot", and PPP-RTK is relative positioning. *Checked.*

Source. `SBASIS-THEORY-E.md` lines 166 to 170.

Answer 2.6 (question, p. 7)

1. $\sum g^i = -2.5$ under $g^1 \equiv 0$; adding $\kappa = 0.625$ to every clock gives $g = (0.625, -2.375, 2.625, -0.875)$ and $a = (3.625, 1.625)$.

2. Pin: $g = (0, -3, 2)$, $a = (3, 1)$; nothing moves. Sum-to-zero over satellites 1 to 3: $\kappa = 1/3$, $g = (0.333, -2.667, 2.333)$, $a = (3.333, 1.333)$: every clock moves by -0.292 m.
3. The pin moves to satellite 2: every clock moves by $+3$ m (the old $\hat{g}^2 = -3$): $g = (\cdot, 0, 5, 1.5)$, $a = (6, 4)$.
4. None: every move is along $\mathbf{1}$, so all differences stay. A user absorbs the common step in its receiver clock at that epoch, but any state it carries across the epoch at the old level sees a step, so the step must be declared and the user resets across it. Sum-to-zero "steps at every constellation-membership change"; a pin moves only when its own satellite leaves, a handover the consumer "pays for with an ambiguity reset". The project uses a pin with a sticky pivot and declares each handover (Question 5.3). *Checked.*

Source. `SBASIS-THEORY-E.md` line 145; `include/network/solve/Sbasis.hpp` lines 162 to 166.

Lesson 3: the real network

Answer 3.1 (question, p. 8)

1. $\boldsymbol{N}_{aa} = 4\boldsymbol{I}_2$, $\boldsymbol{N}_{ag} = -\boldsymbol{J}$, so $\boldsymbol{N}_{ga}\boldsymbol{N}_{aa}^{-1}\boldsymbol{N}_{ag} = \frac{2}{4}\boldsymbol{J}_4$ and $\bar{\boldsymbol{N}}_{gg} = 2\boldsymbol{I}_4 - 0.5\,\boldsymbol{J}_4$ (1.5 on the diagonal, -0.5 off it). With $\boldsymbol{u}_a = (14.5, 6.5)$ (the sums of each receiver's pseudoranges) and $\boldsymbol{u}_g = -(4, 10, 0, 7)$: $\bar{\boldsymbol{u}}_g = \boldsymbol{u}_g + (14.5 + 6.5)/4 = (1.25, -4.75, 5.25, -1.75)$. It sums to zero: the data are consistent along the null direction.
2. $\boldsymbol{J}_4$ has eigenvalues 4 (along $\mathbf{1}$) and 0 (three times), so $\bar{\boldsymbol{N}}_{gg}$ has 0 (along $\mathbf{1}$) and 2 (three times). The null eigenvector is $(0.5, 0.5, 0.5, 0.5)$: weight $1/\sqrt{4}$ on every satellite clock, the clock direction of Question 2.1 seen from the satellites.
3. The rows read $2g^i - 0.5\,S = \bar{u}_i$ with $S = \sum g^i$. With $g^1 = 0$: $S = -2.5$, $g = (0, -3, 2, -1.5)$. Back-substitution $a_r = (u_{a,r} + S)/4$ gives $a = (3, 1)$, the solution of Question 2.2: the elimination is exact. *Checked.*

Source. Equation (5) (status report, step 6); the worked example's data.

Answer 3.2 (question, p. 8)

1. Two. The GPS clock direction: every receiver clock and every satellite clock, GPS and Galileo, $+\kappa$ (each Galileo row holds $a_r - g^i$ too). The Galileo direction: $d_r^E + \kappa_E$ at every receiver and $g^i + \kappa_E$ for every Galileo satellite.
2. Two, spanned by a flat shift of the GPS clocks and a flat shift of the Galileo clocks; the receiver terms are gone. When both eigenvalues are zero an eigensolver may return any rotation of the pair.
3. $1/\sqrt{32} = 0.1768$ on every GPS clock and $1/\sqrt{24} = 0.2041$ on every Galileo clock. A small check (two receivers, three GPS and two Galileo satellites) has 9 unknowns, rank 7 and a two-dimensional reduced null space constant within each constellation. *Checked.*

Source. the arithmetic; the real run in Question 3.5.

Answer 3.3 (question, p. 8)

1. $1 + 2f + (n-1) + m + f(n-1) + f\,m = 1 + 2f + (1+f)(n-1+m)$; with the slant ionosphere $+\,(n-1+m)$.
2. $f = 2$, $m = 32$: $5 + 3(n+31) + (n+31) = 5 + 4(n+31)$, which gives 369 for $n = 60$, 361 for 58 and 1,349 for 305. The slope 4 per station is the figure's "slant ionosphere: 4 per station"; without the slant term it is 3.
3. $1 + 4 + 3 \times 115 = 350$, plus 115: 465. Eq. 24 is a single-system count and the record has no two-system form, so 465 sets the scale only. *Checked.*

Source. `SBASIS-THEORY-E.md` lines 106 to 114; deck slide 6.

Check.

```
C:/Python313/python.exe - <<'EOF'
f = 2
for n, m in ((60, 32), (58, 32), (305, 32), (60, 56)):
    print(n, m, 1 + 2*f + (1 + f)*(n - 1 + m) + (n - 1 + m))
EOF
```

Answer 3.4 (question, p. 9)

1. Shared: satellite clock (GPS, Galileo), satellite phase bias (GPS L1C, L2W; Galileo L1C, L5Q), satellite code bias (GPS C1C, C2W; Galileo C1C, C5Q); the counts (498, 393) are station-held instances, one per line of sight. Local: ambiguity per signal, slant ionosphere per system, zenith wet delay (60), receiver clock (60), Galileo inter-system bias (60), receiver DCB per system; receiver position 180 columns "pinned: held at the roster coordinate". $\mathrm{d}t_r \to$ receiver clock, $\delta t_r \to$ inter-system bias, $T \to$ zenith wet delay, $I \to$ slant ionosphere, $B_{r,s} \to$ receiver DCB, $N \to$ ambiguity; the satellite terms $\to$ the shared rows. The receiver phase bias has no row (the parameter-model line reads `rx_phase_bias no no`, line 380): the ambiguities absorb it, as in Question 1.2.
2. Shared $(1 + 2f)\,m = 5 \times 56 = 280$ (line 639, `nshared=280`). Local: 3 per line of sight (two ambiguities, one slant ionosphere) $\times$ 891 lines of sight, plus 5 per station $\times$ 60: 2,673+300 = 2,973. *Checked.*

Source. `runs/_rf0hv15_sim6/stdout.txt` lines 380, 639, 658 to 686; `src/core/network/assembly/ParamInventory.cpp` lines 645 to 661.

Answer 3.5 (question, p. 9)

1. At tow 259230: $+4.036 \times 10^{-11}$ and -1.803×10^{-10} (a negative value is a rounded zero), with weights 0.2040 on every Galileo clock and 0.1767 on every GPS clock: exactly the flat shifts of Question 3.2.
2. Cutoff $= 10^{-11} \times 1.0459 \times 10^{5} = 1.0459 \times 10^{-6}$; cutoff/$1.803 \times 10^{-10} = 5.8 \times 10^{3}$ (3.8 decades); the weakest real direction, 1.124×10^{-4} (a GPS L2W phase-bias direction), is 107 times the cutoff (2.0 decades). The rule, "cutoff $\geq$ 2 decades above the gauge depth and $\geq$ 2 decades below the weakest-real layer", holds, the upper margin only just. The run end (line 4899) reports 720 epochs at tol 10^{-11}, 720 emitted, none withheld, observed count min 2, max 2.

3. Lumps and rows remove them before the gate, as in Question 2.4. The record maps the types to rows: 1a to the 2 clock pins (acting after the gate); 3a and 3b to 112 code-bias rows; 5 to 112 ambiguity pins; 1b, 2a, 2b and 4 (240) to the receiver side, which is local (free clock, inter-system bias, DCB with 120 prior entries, no receiver phase-bias state). $2+112+112+240 = 466 = 465+1$, the extra one being the second constellation's clock direction. *Checked.*

Source. `rf_null_space_read.py` output; `runs/_rf0hv15_sim6/logs/network_estimability.csv` rows 2 to 6; `runs/_rf0hv15_sim6/stdout.txt` lines 640, 4899; `RankGate.cpp` lines 408 to 410; `SBASIS-CONTRACT.md` lines 59 to 68.

Lesson 4: the project's reference choices

Answer 4.1 (question, p. 10)

1. An estimable function $\boldsymbol{f}^{\mathrm{T}}\boldsymbol{x}$ ($\boldsymbol{f}$ in the range of $\boldsymbol{A}^{\mathrm{T}}$) "is invariant in value AND variance under any admissible datum", that is, under every reference choice that removes exactly the null space. Question 2.2 showed the value half: four differences equal in two choices.
2. Position, zenith delay and vertical ionosphere are estimable before any choice (their rows of the null-space basis are zero). A satellite clock and a satellite phase bias are not: "The satellite clock does not exist as an estimable number; only a datum-dependent lump does."
3. The differential wander was estimable, so no reference choice could move it, and every datum comparison came out negative: "The cure must alter ESTIMATION (weights / dynamics / model / transmitted covariance), not the gauge." The coverage change that cut the wander from 47 to 14 cm was such a change.

Source. `SBASIS-THEORY-E.md` lines 40 to 41, 75 to 83, 116 to 120, 223 to 225.

Check.

```
sed -n '75,83p;116,120p' docs/cards/network-vehicle/SBASIS-THEORY-E.md
```

Answer 4.2 (question, p. 11)

The verify prints, per reference configuration, "76 recommended knobs armed in regime, 2 unarmed (2 documented exceptions), 0 failure(s)" and "golden verify: PASS". *Checked.*

1. Clock pin: no key (the default), with `sticky_pivot` (line 324) and `handover_stransform` (343). Anchors: `phase_datum_prioritized` (297), `phase_datum_held_basis` (925), `ambiguity_vehicle float_kf` (924). Code-bias datum: `estimate_sat_code_bias` (293), `sat_gf_datum broadcast_tgd` (922); the ionosphere-free pin and sum-to-zero rows have no key of their own. Receiver DCB prior: `rx_code_bias_datum` (292).
2. The pin is unaffected by satellites rising and setting until the pivot leaves; then the constellation is re-referenced (one handover per constellation in 6 h: E02 to E03, G01 to G02). An anchor is a single point of failure: a handover steps the bias by whole cycles unless the successor is pinned at its own level (308 basis values elected, 22 of them cold, declared). The sum-to-zero rows would move at every rise and set, as in Question 2.6; the simulated case suppresses them because the pins already fix every channel (2,880 suppressed $= 4 \times 720$). The ensemble was "SUPERSEDED BY MEASUREMENT" and deleted (commit 7914263).

3. `clock_datum_sum_to_zero`, `clock_datum_broadcast_anchored`, `pivot_election`; `phase_datum_eq34` is false (line 345).

Source. `SBASIS-THEORY-E.md` lines 143 to 150, 239 to 259; `Sbasis.hpp` lines 162 to 166; `config/golden/golden_sim_refound.json`; `_rap_scb_g_q6/stdout.txt` lines 4385, 4400, 4845.

Check.

```
C:/Python313/python.exe scripts/make_golden_config.py --verify
grep -n -e '"sticky_pivot"' -e '"phase_datum_held_basis"' -e '"phase_datum_eq34"' \
    config/golden/golden_sim_refound.json
```

Answer 4.3 (question, p. 11)

1. $56\times(1+2+2) = 280$ shared columns. Clock pins 2 (E02, G01). Ambiguity pins $112 = 32\times2+24\times2$, one per phase-bias column, on 18 stations. Code-bias rows 112: 56 ionosphere-free pins and 56 geometry-free pins at T_{GD} (54 at cT_{GD}, 2 with a broadcast T_{GD} of 0); sum-to-zero 0. Receiver prior entries $120 = 60 \times 2$, each of information $1/(10\text{ m})^2 = 0.01\text{ m}^{-2}$. Total: 226 rows on the shared state plus 120 prior entries.
2. The ambiguity pins and code-bias rows act before the gate and fix the 224 phase-bias and code-bias columns; the 56 clocks keep one flat direction per constellation (Question 3.2); the clock pins act in the solve after the gate, so the gate expects 2 (`Sbasis.hpp` lines 50 to 55: "exactly #constellations residual deficiencies").
3. On a mismatch the gate throws; the heal soft-anchors near-null phase-bias columns (σ 1 m) and re-gates once; if the count still differs the epoch is withheld and counted as under- or over-specified. Completion would first fold one eigenvector row per extra separable near-null direction; it folded none in 720 epochs (line 4112). *Checked.*

Source. `_rap_scb_g_q6/stdout.txt` lines 631, 679 to 738, 4109, 4112, 4387; `Sbasis.hpp` lines 50 to 55; `src/core/network/solve/RankGate.cpp` lines 193 to 305.

Check.

```
sed -n '631p;679,738p;4109,4112p' \
    D:/GnssProcessorSupport/runs_nom/_rap_scb_g_q6/stdout.txt
```

Answer 4.4 (question, p. 11)

1. `B,G05,C1C,-3.2108400641001413` and `B,G05,C2W,-5.288075205569585`: the ratio is $1.646944 = \mu_2$, so the ionosphere-free combination is zero. The navigation record gives $T_{GD} = -1.071020960808 \times 10^{-8}$ s, and $cT_{GD} = -3.2108400641$ m, the C1C record. The pinned pair carries no estimate: its ionosphere-free part is pinned to zero and its geometry-free part to the broadcast group delay (54 satellites at cT_{GD}, 2 with a broadcast value of 0; stdout line 631).
2. With a slant ionosphere the satellite's true geometry-free code bias is not estimable: it merges into the slant ionosphere of every line of sight to that satellite (Question 1.3, mirrored at the satellite). The corrections carry no ionosphere record, so a user estimates its own slant ionosphere, which absorbs the same lump: code biases and ionosphere act as a pair. *Checked.*

Source. `_rap_scb_g_q6/products.txt` lines 61 to 62; `data/2024-129/products/BRDC00IGS_R_20241290000_01D_MN.rnx` line 568; `docs/refound/RF-0c_gf_pin_2026-09-15.md` lines 22 to 23; `SBASIS-CONTRACT.md` lines 114 to 133.

Check.

```
C:/Python313/python.exe - <<'EOF'
print(299792458 * -1.071020960808e-08)          # c * T_GD of G05, m
print(-5.288075205569585 / -3.2108400641001413)  # C2W / C1C
print((1575.42 / 1227.60) ** 2)                 # mu_2
EOF
```

Answer 4.5 (question, p. 12)

1. $0.0074/10^2 = 7.4 \times 10^{-5}$ m^{-2} along the unit level; $\sigma = 1/\sqrt{7.4 \times 10^{-5}} = 116$ m.
2. $(0.0074 + 0.2392^2 + 0.1334^2)/0.0074 = (0.0074 + 0.0572 + 0.0178)/0.0074 = 11.1$, σ about 35 m, still useless: card R1D found such a row changes nothing at 10 m, and at 1 m shrinks the move only by a fifth while pushing 0.95 m into C1W rows: "the datum row is not the fix".
3. $\sigma = 1/\sqrt{110} = 0.095$ m. The carried ambiguity coupling supplies 109 of the 110, the DCB prior 7×10^{-5}: the coupling to the shared state holds the level, not the datum.
4. The direction lives inside one station's block, which is eliminated before the shared matrix is formed (Question 3.1), so the gate never sees it. MAJU00MHL anchored phase biases of five GPS satellites (13 pins). With the pins held, the station's level plus a matching shift of its anchored phase biases is invisible to its data (residual 2.8×10^{-11}), so a move of the level reaches the published phase biases. In the GPS L5 event the level moved -21.92 m; barring the station from the anchor election removed the event: "The anchor loop is the transmission". The hold card ALD derives is not built; GPS L5 stays sequenced.

Source. card R1D lines 21 to 28, 54, 66 to 89, 99 to 122, 159; card ALD lines 27, 31 to 41, 54, 58 to 65.

Check.

```
C:/Python313/python.exe - <<'EOF'
import math
print(1 / math.sqrt(0.0074 / 100))                  # sigma from the prior, m
print((0.0074 + 0.2392**2 + 0.1334**2) / 0.0074)     # information factor
print(1 / math.sqrt(110))                           # sigma from the coupling, m
EOF
```

Lesson 5: what a user sees

Answer 5.1 (question, p. 12)

1. Δ enters every code and phase row of system g identically; the receiver clock (with the inter-system bias for a second system) takes it up, as $\hat{a}_C$ took up a_A in Question 2.5. The geometry, hence the position, is unchanged.
2. A constant d enters only that satellite's phase rows, as a constant: the float ambiguity moves, $N \to N + d/\lambda$. The position is unchanged; only the ambiguity's meaning changes, which matters once a user fixes integers.

3. $(\delta, \mu_2\delta)$ adds $\mu_s\delta$ to code row s, the ionosphere's own signature, so $I^i \to I^i + \delta$ absorbs it; on phase that shift enters as $-\mu_s\delta$, which the float ambiguities absorb. The position is unchanged; the user's slant ionosphere is biased by δ. The ionosphere-free combination of $(\delta, \mu_2\delta)$ is zero, so an ionosphere-free user sees nothing at all.
4. The per-system mean at each epoch (case 1, the receiver clock), then each satellite's mean over the session (case 2, the ambiguity). Case 3 does not arise for clocks. The S-system statement: the user uses only the block-diagonal part of the correction covariance, "the rank-one $[e; e]$ (clock/geometry) and $[-\mu; \mu]$ (iono) parts lie in the user's own design range and are absorbed".

Source. `SBASIS-THEORY-E.md` lines 158 to 182; `SBASIS-CONTRACT.md` lines 98 to 102, 114 to 133; status report, step 5; `results/t3/results.md` lines 64 to 66.

Answer 5.2 (question, p. 13)

Field	Declares	Defined or written at
`S30000SIM`	the run's reference (clock-datum) station; the pins act on satellite clocks	`ApplicationNetworkSetup.cpp` line 1485
`1`, `nominal-wu93`	bias sign convention; phase wind-up model	`Sbasis.hpp` lines 98 to 108, 215
`0` (5th field)	ambiguity datum: "legacy single pivot and prioritized anchor"	`Sbasis.hpp` lines 129 to 132
`0` (6th field)	clock datum: one hard satellite clock pin per constellation	`Sbasis.hpp` lines 162 to 166
`1`, `0`	the network emits its own code biases; code-bias datum id	`Sbasis.hpp` lines 229, 238
`0.05`, `1`	clock discontinuity threshold (m); discontinuities emitted	`NetworkProductEmitter.hpp` lines 195 to 207
the SP3 and CLK names	the a priori orbit and clock; every C record corrects this clock	`ApplicationNetworkSetup.cpp` lines 1524 to 1566
`E:E02`, `G:G01`	realized clock pivots (C = 0 at the first epoch)	`SolPrep.cpp` lines 1176 to 1179
`cbjump`, `atx`, `oref`	code-bias jump test; antenna file; orbits at the centre of mass	`NetworkProductEmitter.hpp` lines 222 to 228; `NetworkProductIO.hpp` lines 270 to 273
`discfrac`, `hsig`	phase jump gate of 0.25 cycle; σ_0 of the H records	`NetworkProductEmitter.hpp` lines 238 to 248; `NetworkProductIO.hpp` lines 238 to 247
`cbpin=...`	the ionosphere-free pin pair per system	`SolPrep.cpp` lines 1118 to 1157
`sat_gf_datum=broadcast_tgd`	the geometry-free datum at cT_{GD}	`SolPrep.cpp` lines 1165 to 1172
`pbhold=2`	the phase-clock hold (Question 6.3)	`ApplicationNetworkSetup.cpp` lines 1599 to 1601

The reference choices of Section 4 are the two datum ids, the pivots, the `cbpin` pairs and `sat_gf_datum`; `pbhold` declares how the phase biases are held. A current run appends `wumode=accumulated`. The real-data S line declares only `cbpin=G:C1C-C2W`: the Galileo pairs are "NOT declared (per-satellite split)" (`runs/_rf0hv15_field6/stdout.txt` line 686).

Source. `src/core/products/NetworkProductIO.cpp` lines 465 to 602 and the lines in the table (paths under `src/core/network/` and `include/`).

Check.

```
sed -n '2p' D:/GnssProcessorSupport/runs_nom/_rap_scb_g_q6/products.txt
```

Answer 5.3 (question, p. 13)

1. The awk prints line 134213, `E,2313,271230.00000715256,400,0,0,1,E` (Galileo, 03:20:30), and line 189631, `E,2313,276149.99999547005,564,0,0,1,G` (GPS, 04:42:30). The script prints "31 satellites, mean step 1.6278 m, sd after the mean 0.0027 m"; an ordinary pair of epochs (276090 to 276120) gives −0.0001 and 0.0008 m.
2. The handover of Question 2.6 (3): the pivot moved from G01 to G02 and every GPS clock stepped by the old pivot-relative value. A user's receiver clock absorbs the common 1.63 m; 3 mm per satellite stays visible.
3. A raw-step grade sees 31 satellites jump 1.6 m at once; the datum-invariant step is 3 mm. On real data the GPS group mean steps by about +154 cm at its pivot change. *Checked.*

Source. `runs/_rf0hv15_sim6/products.txt`; `results/t3/results.md` lines 105 to 107.

Answer 5.4 (question, p. 14)

1. A chord is $e_i = s_i - s_{\text{ref}}$: $\text{Var}(e_i) = \sigma^2 + \sigma_{\text{ref}}^2$ and $\text{Cov}(e_i, e_k) = \sigma_{\text{ref}}^2$, so two chords correlate by $\sigma_{\text{ref}}^2/(\sigma^2 + \sigma_{\text{ref}}^2)$, 0.5 when $\sigma_{\text{ref}} = \sigma$.
2. L1C over all epochs: as traced, median $|e|$ 0.304 cycle, 65.3 % round to the right integer; reference-free 0.122 cycle, 81.6 %. Last quarter: 0.260 cycle and 69.5 % against 0.092 cycle and 86.5 %.
3. The reference satellite's error and its returns enter every chord, so a grade on the traced chords reports that one element's share: "Quote DD grades ref-free from now on." A second trap of the same card: subtracting a group's own circular median shrinks uniform noise from 0.251 to 0.145 cycle, so a wrapped grade is quoted as a gap to a size-matched null. *Checked.*

Source. `docs/cards/sim-bed/DD-TRACE-REF-SAT.md` lines 41 to 48, 107 to 110, 143 to 159; `CLAUDE.md`, "Grading traps that still apply".

Lesson 6: the satellite clock

Answer 6.1 (question, p. 14)

1. $\sqrt{2.9 \times 10^{-8} \times 30} = 0.933$ mm per epoch; $\sqrt{2.9 \times 10^{-8} \times 21{,}600} = 2.50$ cm over 6 h. The value is measured: "$q_{\text{phys}} \approx 2.9 \times 10^{-8}$ m^2/s → 0.94 mm per 30 s".
2. The drift after n epochs is $0.30n$ mm, the walk's 1σ $0.933\sqrt{n}$ mm; they are equal at $n = (0.933/0.30)^2 = 9.7$ epochs, about 5 min. At 6 h the drift is 21.6 cm, 8.6 times the walk. A step standard deviation is being compared with a signed, persistent increment: a drift grows as n, a random walk's spread as $\sqrt{n}$, so a walk of this density cannot follow a sustained rate. Card SCB reads the outcome as partial tracking.

Source. `config/golden/golden_sim_refound.json` line 323; `include/network/solve/ClockAnchor.hpp` lines 35 to 36; `docs/cards/clock/Q3-PHYSICAL-FLOOR.md` lines 29 to 32; card SCB lines 29 to 35.

Answer 6.2 (question, p. 15)

1. $R = 0.0923^2/2 = 4.26 \times 10^{-3}$ m^2; $K = \sqrt{8.7 \times 10^{-7}/4.26 \times 10^{-3}} = 0.0143$; $\Delta t/K \approx 2{,}100$ s, about 35 min (the exact steady state of the scalar filter gives 2,084 s).

2. $10^{-5} \times 2{,}100 \approx 2.1$ cm. Card SCB measured 2,127 s (2,000 s GPS only): the anchor's q sets the time constant, within about 2 %. Two cautions: the argument is scalar and stationary, and the real filter is neither; and the grader's error is estimate minus truth with a positive coefficient, which a pure lag would make negative. The card calls it a lag and does not explain the sign, so the safe wording is "an error proportional to the drift rate, with a coefficient of about 35 min".
3. $K = \sqrt{1.5 \times 10^{-4} / 4.26 \times 10^{-3}} = 0.19$, about 160 s (exact steady state 146 s), the scale of the measured 52 s: the regime's larger q follows the drift within the epoch noise. *Checked (arithmetic).*

Source. `docs/cards/clock/Q3-ANCHOR-DESIGN.md` lines 47 to 56; card SCB lines 124, 128; `docs/cards/raim-qc/DAY129-FRAME-ANATOMY.md` lines 139 to 143; `scripts/simnet_clock_truth_grade.py` line 151.

Check.

```
C:/Python313/python.exe - <<'EOF'
import math
R = 0.0923**2 / 2
for q in (2.9e-8, 5e-6):
    K = math.sqrt(q * 30 / R)
    print(q, 'K %.4f' % K, 'tau %.0f s' % (30 / K), 'error %.4f m' % (1e-5 * 30 / K))
EOF
```

Answer 6.3 (question, p. 15)

1. $C - P = c(t) - p(t_0) - [\tilde{c}(t) - \tilde{c}(t_0)] = (c - p)(t_0) + M(t) - M(t_0)$: constant per satellite up to the system mean M, which the receiver clock absorbs. On code the user applies $a(t) + C(t)$, the network's estimate; on phase it sees $a(t) + C(t) - P(t) = a(t)$ plus a constant: the a priori clock's motion. With truth $a(t) + e(t)$, the phase rows are wrong by $-[e(t) - e(t_0)]$ plus a constant; the float ambiguity absorbs the constant, the change passes one to one.
2. Mean square over hour 6 (t from 18,000 to 21,600 s): rate part $\sigma_r^2 (t_{\mathrm{hi}}^3 - t_{\mathrm{lo}}^3)/(3 \times 3600) = 10^{-10} \times 3.931 \times 10^8 = 0.0393$ m^2; walk part $q(t_{\mathrm{lo}} + t_{\mathrm{hi}})/2 = 0.0006$ m^2; rms 0.200 m, against the measured 0.201 m (GPS) and 0.206 m (Galileo).
3. The loader refuses `network.emit_pb_hold_phase_clock` with `network.apriori_source sp3_brdc_clock`: "the phase-clock hold is valid only with a precise a priori clock product"; with `brdc_rts` it only warns (lines 2612 to 2617). Part 1 is the reason: the hold hands the user the a priori clock's motion, so the a priori clock must be precise. The S line declares `pbhold=2` (3 for the smoothed hold of the self-contained postures). On real data, ignoring the precondition cost the user 1.191 and 0.743 m in hours 1 and 2, against 0.108 and 0.024 m with CODE. *Checked.*

Source. `include/network/emit/PbHold.hpp` lines 60 to 66; card LAH lines 30 to 36; card SCB lines 129 to 145; `src/core/config/GnssConfigLoadNetwork.cpp` lines 311 to 319, 2612 to 2617; card SCL lines 22 to 26, 76 to 78.

Check.

```
C:/Python313/python.exe - <<'EOF'
lo, hi = 18000.0, 21600.0                       # hour 6, s after the reset
rate = 1e-10 * (hi**3 - lo**3) / (3 * 3600)     # rate part, m^2
walk = 3e-8 * (lo + hi) / 2                     # random-walk part, m^2
print((rate + walk) ** 0.5)
EOF
sed -n '311,319p' src/core/config/GnssConfigLoadNetwork.cpp
```

Answer 6.4 (question, p. 16)

Reference case `_rap_scb_g_k6`: clock level at hour 6 0.0533 m; per-satellite mean over hours 2 to 6, rms 0.0426 m over 55 satellites; correlation with the drawn rate +0.524; user by hour 0.1976, 0.2012, 0.3189, 0.2142, 0.2370, 0.3376 m. Self-contained regime `_rap_scb_c_k6`: 0.0232 m; 0.0205 m over 31 satellites; +0.026; user 0.2287, 0.1296, 0.1003, 0.0211, 0.0311, 0.0260 m. The regime follows the drift (no correlation with the rate); the reference case does not. Its user error is about ten times its clock error because the hold passes the a priori clock's error to the phase rows whatever the estimator tracks (Question 6.3). Not shown: one realization of an error with chosen statistics; the sign question of Question 6.2; the regime case is GPS only (31 satellites against 55); neither the regime nor the native posture is armed. Card SCB's text says the regime and native users "do not move" between the two data sets; its tables show the native user up to 2.1 cm higher on the clock data set in hours 1 to 3. *Checked.*

Source. the two grader outputs; card SCB lines 8, 10, 122 to 145.

Answer 6.5 (question, p. 16)

1. $0.0024 - 0.0007 + 0.0099 = 0.0116$ m, not 0.0171 m: the steps add exactly for each station-day, but the median of a sum is not the sum of the medians. The clock's information is the broadcast-clock step, about 1.0 cm.
2. The native posture against the GPS-only reference configuration of 2026-10-07 (GPS and Galileo since 2026-10-13), days 130 to 132 (9 to 11 May 2024), 16 stations, as the median paired change of each station's hours 2 to 6 error; 19 of 48 station-days (39.6 %) are worse by more than 2 cm; day 129 is its most favourable day (+0.3 cm); the first hour adds +7.65 cm.

Source. card NBP lines 170 to 177, 184 to 190, 210 to 219; card GAD.

Lesson 7: ambiguities

Answer 7.1 (question, p. 17)

$\lambda_1 = c/f_1 = 0.190294$ m, $\lambda_2 = 0.244210$ m, $\mu_2 = (77/60)^2 = 1.646944$. $\cos\theta = (\lambda_1 + \mu_2\lambda_2)/(|(1,\mu_2)|\,|(\lambda_1,\lambda_2)|) = 0.592494/(1.926765 \times 0.309597) = 0.99325$, $\theta = 6.66°$. The part orthogonal to the ionosphere is $0.3096 \times \sin 6.66° = 0.0359$ m of the 0.31 m step. "99.3 %" is the cosine, a share of amplitude; as a share of variance ($\cos^2$) it is 98.7 %. The metric assumes equal phase noise in metres on both bands. *Checked.*

Source. card ARL line 43 (the statement); the arithmetic.

Check.

```
C:/Python313/python.exe - <<'EOF'
import math
c = 299792458.0
l1, l2 = c / 1575.42e6, c / 1227.60e6
mu2 = (1575.42 / 1227.60) ** 2
k = (l1 + mu2 * l2) / math.hypot(1, mu2) / math.hypot(l1, l2)
print(k, math.degrees(math.acos(k)), math.hypot(l1, l2) * math.sin(math.acos(k)), k * k)
EOF
```

Answer 7.2 (question, p. 17)

1. Cycles per metre of L1 ionosphere: L1 $1/\lambda_1 = 5.255$, L2 $\mu_2/\lambda_2 = 6.744$, wide lane $5.255 - 6.744 = -1.489$ (the card: "L1C 5.255, L2W 6.744"). $d = 0.4233$ m: L1 2.22 cycles, wide lane 0.63 cycle. $d = 0.0419$ m: L1 0.22 cycle, wide lane 0.062 cycle.
2. 0.1 L1 cycle = 0.0190 m; 1 TECU = $40.3 \times 10^{16}/f_1^2 = 0.1624$ m of L1 delay, so 0.1 cycle is 0.117 TECU. The no-pin error, 0.4233 m = 2.61 TECU, is 22 times that: "2.61 TECU, 22× the 0.117 TECU that moves a fraction". The 0.4233 m is a per-line-of-sight rms, so a double difference of several such terms can be larger. *Checked.*

Source. `docs/cards/raim-qc/T3-IONO-MECHANISM.md` lines 326 to 331; `docs/SPEC_oracle_pin.md` lines 514 to 517, 541 to 546.

Answer 7.3 (question, p. 17)

1. The pins make the network emit $\bar{b}^i = b^i + N_A^i - \zeta^i$ (in cycles), and the user then estimates $\tilde{N}_C^i = N_C^i - N_A^i + \zeta^i = (z_C^i - z_A^i) + (\beta_C - \beta_A) + \zeta^i$.
2. $\tilde{N}_C^i - \tilde{N}_C^k = [(z_C^i - z_C^k) - (z_A^i - z_A^k)] + (\zeta^i - \zeta^k)$: the bracket is an integer double difference (the receiver phase biases cancel), so the difference is an integer exactly when $\zeta^i - \zeta^k$ is. Integer pins (Question 2.5: $\zeta = 0$) give integers; float pins give a fraction.
3. "The WIRE's gauge is an arbitrary receiver's FLOAT ambiguity": the anchors are float targets, re-pinned at the successor's float on every handover, so the user's between-satellite differences carry fractions spread over a whole cycle, and the published biases sit at the uniform null (Question 7.5). Card IRD: a user's single difference is integer if and only if every emitted bias difference is; it proposes to pin a spanning tree of each observation graph ($n_c + m_c - 1$ edges) to integers plus one real-valued root receiver phase-bias pin, so that every chord is an integer (after de Jonge [1]). Not built (IRD-1 is the next step).

Source. `docs/network_ambiguity_resolution_design.md` lines 335, 352 to 356; card IRD lines 19, 41 to 49.

Answer 7.4 (question, p. 18)

Row 1b, the double-difference-invariant per-line-of-sight L1 slant ionosphere error, carries the deficit: 0.4233 m with no pin, 0.0419 m with every ambiguity pinned (10.1 times smaller; 2.61 to 0.26 TECU). "The `none` column's row 1b IS the integer-information deficit". The raw per-line-of-sight error (2.73 m) is about 97 % station and satellite lumps, which the reference choice and the user absorb, so it cannot measure integer information. Partial pins do not interpolate: L5Q alone 0.2793 m (better); L1C with L2W 0.5632 m (worse than no pin, although every double difference is then integer); one station 0.4476 m; one satellite 0.4741 m. "Integer information is worth having only when it is complete on the LOS." The proposed cause (the bands of a line of sight share one ionosphere state, so exact integers on a subset over-determine it) is called a consistent reading, not a proof. The numbers come from a 60-epoch warm resume on `simnet60_quiet_wu93_24h`, a data set whose numbers `CLAUDE.md` marks superseded: the reading stands, the values are not comparable with today's simulated data.

Source. `docs/SPEC_oracle_pin.md` lines 20 to 29, 193 to 195, 514 to 517, 534 to 546, 558 to 568, 611 to 627, 654 to 663.

Answer 7.5 (question, p. 18)

First command (4 s): the published GPS phase biases' wide-lane median |fraction| by hour from 00:30:15 is 0.254, 0.260, 0.259, 0.255, 0.253, 0.252, the narrow lane 0.264 to 0.266, against a uniform null of 0.250 (band about 0.22 to 0.27). Second (4 s): "CORRECT-FIX RATE among kept:

5572/5572 = 100.00%", "WRONG FIXES: none". The integers are in the network's float solution (a search on its float states and covariance fixes 5,572 of 13,138 settled double differences of that case, all correct), but the published phase biases carry none of it: they sit at the uniform null, as Question 7.3 predicts for float pins. A user cannot fix integers from these corrections. The 42 % case is a simpler 2 h simulation without time-correlated phase errors or slips; with the time-correlated phase error the share is 18.5 %. *Checked.*

Source. the two outputs; card ARL lines 98 to 102, 142.

Lesson 8: simulated data

Answer 8.1 (question, p. 19)

The quiet data set's truth clock is the CODE final clock, and the network takes the same product as its a priori clock, so the estimable satellite clock truth is identically zero (card SCB, section 1). `results/t2/results.md` confirms it: no `sat_clock` key in the generator's configurations, no ARMED line, no `truth_satclock.csv`. The two `ls` commands print 0 and 61. The 0.9 cm agreement therefore "certifies the algebra and the biases, not the clock": a simulation grades what it injects, plus contrasts in which every reused term cancels. *Checked.*

Source. card SCB lines 17 to 25; `results/t2/results.md` lines 123 to 126; `docs/cards/sim-bed/SIMNET-REF.md` lines 269 to 272.

Answer 8.2 (question, p. 19)

1. 0.504167 h = 1,815.0012 s; epoch k is $30k$ s after the first, and the first k with $30k \geq 1{,}815.001$ is 61 (1,830 s): tow 261060, 00:31:00. 0.4958333333 h = 1,785 s gives $k = 60$: tow 261030, 00:30:30 (the T2 restart). Each offset sits 15 s, half an epoch, below the epoch it selects, so rounding cannot move the start by an epoch. The grading origin 00:30:15 (tow 261015) is not a start epoch: the reference case starts at 00:00:30 and is graded from 00:30:15; a case with the offset starts at 00:31:00.
2. Output: 259230 367, 259260 366, 259290 366, 259320 367, 259350 309 (warnings per epoch): 5 epochs logged. A warning needs more than 150 s outside the window, which opens at tow 259500, so 259380 to 259470 are extrapolated without one: 9 epochs, 00:00:30 to 00:04:30. "Faulty in the first 30 minutes" described the grading origin, not the fault; what lasts is the state those epochs leave (Question 8.3). The report now says 9 epochs. *Checked.*

Source. `include/math/TimeUtils.hpp` lines 416 to 448; `docs/cards/queues/QPR-ITEMS-5-8.md` lines 239, 256; `src/core/model/ModelGeometry.cpp` line 1204; `docs/refound/CTU2_level_recursion_followup_2026-09-29.md` lines 29 to 43.

Check.

```
head -2 D:/GnssProcessorSupport/runs_nom/_rap_scb_g_q6/logs/closed_loop_trace.csv
```

Answer 8.3 (question, p. 19)

Baseline (from 00:00:30): level by hour 0.7087, 0.3487, 0.3303, 0.3157, 0.3096, 0.3108 m; per-satellite mean over hours 2 to 6, rms 0.3179 m over 55 satellites (G06 −1.198, E05 +0.872, E36 −0.846). Card SCB's quiet case (from 00:31:00): level 0.0809, 0.0291, 0.0226, 0.0202, 0.0193, 0.0185 m; rms 0.0186 m. With `--no-ds` the baseline's rms reads 0.7784 m. The datum share is the ionosphere-free combination of the truth code biases over the pinned pair: the network's

clock contains it by construction of the code-bias datum (Question 4.4), so it belongs to the truth expressed in the network's basis. Truth is transformed; differencing the estimates instead would bias the variance factor downward, because the estimates share the datum. The baseline's user does not show the 0.32 m: a float user cannot see a per-satellite constant (Question 5.1). *Checked.*

Source. the grader outputs; card SCB lines 87, 102, 107 to 109; `scripts/simnet_clock_truth_grade.py` lines 138 to 153; `SIMNET-REF.md` lines 276 to 283.

Answer 8.4 (question, p. 20)

Ours minus perfect, in metres: +0.0030, +0.0026, +0.0085, −0.0007, +0.0005, +0.0002. The largest cost is 0.85 cm, in hour 3 (the report's "at most 0.9 cm"); hours 4 to 6 show nothing measurable. The floor (perfect corrections, hours 4 to 6) is 1.73, 1.81 and 1.76 cm, in the height: median $|U|$ 1.54, 1.54, 1.65 cm against $|E| \leq 0.46$ cm and $|N| \leq 0.55$ cm. Hour 6 is 30 min long, because the network run ends at 06:00.

Source. `results/t2/results.md` lines 195 to 206; `results/t2/metrics.csv`.

Lesson 9: real data

Answer 9.1 (question, p. 20)

`rapid_arm.py score`: `_rf0hv15_field6` median by hour 0.06, 0.01, 0.02, 0.01, 0.02, 0.01 m, 688 epochs below 10 cm; `_rf0hv15_sim6` 0.14, 0.07, 0.03, 0.02, 0.03, 0.02 m, 595. `gcensus.py`: `galp` (GPS and Galileo corrections), 16 stations, median by hour 0.136, 0.054, 0.041, 0.036, 0.040, 0.037 m, WTZZ 0.058, 0.011, 0.018, 0.010, 0.015, 0.011 m; paired against `galpc` (GPS only), hours 2 to 6 median −0.0198 m, 10 better, 2 worse (HARB +0.0158, DAV1 +0.0605), hour 1 −0.2203 m; `galpg` (GPS-only users of the GPS and Galileo corrections) −0.0000 m, ARHT +0.1958 m. Better and worse mean a change of more than 1 cm. The first hour: WTZZ00DEU 5.8 against a mean of 1.3 cm over hours 2 to 6 (4.5 times); the median station 13.6 against 4.2 cm (3.3 times). Mode: `position_q` 100 (moving receiver), `ar_mode` 0 (no integer fixing). None of the 17 user stations is in the 58-station network; WTZZ00DEU shares its site with network stations WTZR00DEU and WTZS00DEU ("the colocated consumer hold-out"; its RINEX header position is about 3 m from WTZR00DEU's). Whether that explains its rank has not been tested. *Checked.*

Source. the outputs; `runs/_rf0hv15_field6/arm.json`; `scripts/clock_level/gcensus.py` lines 151 to 153; card GAD lines 70 to 75; `docs/refound/RF-PD_station_rejection_2026-09-15.md` line 87.

Check.

```
C:/Python313/python.exe - <<'EOF'
import json
cfg = json.load(open('config/golden/golden_field_refound.json'))
names = [s['name'] if isinstance(s, dict) else s for s in cfg['network']['stations']]
print(len(names), [n for n in names if n.startswith('WTZ')])
EOF
```

Answer 9.2 (question, p. 21)

Hours 2 to 6 of days 128 to 132: the 25 sixteen-station medians have a median of 0.038 m and a mean of 0.0406 m; the 25 WTZZ00DEU values a median of 0.018 m. On day 130 WTZZ00DEU reads 3.9, 5.9 and 5.3 cm in hours 4 to 6 against the median station's 3.5, 3.2 and 3.8 cm: "the best station" holds on 8 May 2024 (day 129), not on every day. The report, which first quoted 3.9 cm for the 16 stations, now quotes 3.8 cm (mean 4.1 cm). *Checked.*

Source. `D:/GnssProcessorSupport/runs_nom/mday_galileo_grades.txt`.

Check.

```
C:/Python313/python.exe - <<'EOF'
import re, statistics as st
cen, wtz = [], []
for l in open('D:/GnssProcessorSupport/runs_nom/mday_galileo_grades.txt'):
    m = re.match(r'== mdg\d+c: .*by hour ([\d. ]+); WTZZ ([\d. ]+)', l)
    if m:
        cen += [float(x) for x in m.group(1).split()[1:]]
        wtz += [float(x) for x in m.group(2).split()[1:]]
print(len(cen), st.median(cen), round(st.mean(cen), 4), st.median(wtz))
EOF
```

Answer 9.3 (question, p. 21)

Against the GFZ final clock, after the per-system epoch mean and each satellite's mean are removed (STD): our network (CODE final plus c0) GPS 4.3 and Galileo 7.6 cm over 01:00 to 06:00 (6.4 and 11.6 cm including the first hour; Galileo RMS 15.8 and 25.3 cm because of E12); CODE final 1.4 and 2.6 cm over 00:00 to 06:00, 1.2 and 2.2 cm over 01:00 to 06:00, 1.9 and 3.3 cm over 13:24 to 24:00; the IGS real-time service 4.2 and 3.2 cm over 13:24 to 24:00, the only hours archived. GFZ is the reference because the network's a priori clock is CODE's: against CODE the network would be measured partly against itself. Over the same hours the corrected clock is about 3.5 times noisier than CODE alone for GPS. The correlation of c0 with (GFZ minus CODE) is GPS −0.08, Galileo +0.01, and with (ESA minus CODE) GPS −0.18, Galileo +0.01: c0 carries no information about CODE's error; relative to its a priori it behaves as added noise of 6.3 cm (GPS). A clock comparison alone does not isolate what a user obtains: the user's phase term is phase bias minus clock, which the hold keeps at CODE's clock (Question 6.3). *Checked.*

Source. `results/t3/table_clock_report.tex`; `results/t3/results.md` lines 9 to 14, 100 to 113; `results/t3/clock_stats.csv`, rows `COD,GFZ,01-06`.

Check.

```
T3=D:/GnssProcessorSupport/presentation/status_update_2026-10-29/results/t3
grep "COD,GFZ,01-06" $T3/clock_stats.csv
```

Answer 9.4 (question, p. 22)

(a) *Model and check.* May: user and network use one undifferenced, uncombined observation model, and the user one filter in every step; the rank gate found exactly the declared 2 directions (one clock per constellation) on 720 of 720 epochs, none withheld, on both reference cases (Question 3.5). May not: that the network uses the user's filter (it has its own information-form solve); that the check proves the choices harmless (it counts null directions only); that it covers each station's level (Question 4.5).

(b) *Reference choices.* May: they are declared in the correction file and every change of pivot is marked (Questions 5.2 and 5.3). May not: "a few" (226 rows and 120 prior entries at the first simulated epoch, Question 4.3); all declared on real data (the Galileo pin pairs are not); that they keep a user's integers (Question 7.3).

(c) *Satellite clock.* May: its precision comes from CODE's final clock; the network's correction adds code noise and no clock information (Question 9.3); on simulated clock error the self-contained postures reach about 2 cm, unarmed (Question 6.4). May not: that it beats real-time clocks; that the network is free of outside clock products; that the quiet simulation tests the clock (Question 8.1). Without CODE's clock a user was 1.7 cm worse at the median, against an older GPS-only configuration (Question 6.5).

(d) *Integers.* May: on a simplified simulation a log-only search fixed 42 % of 13,138 double differences, none wrongly (Question 7.5). May not: integers in the corrections; integers on real data; producer-side fixing, whose standing verdict is DO NOT BUILD (`docs/network_ambiguity_resolution_design.md` lines 3 to 5).

(e) *Results.* May: on 8 May 2024, WTZZ00DEU 1.0 to 1.8 cm and the median of 16 further stations 3.6 to 5.4 cm in hours 2 to 6, without integers; over five days the median station 3.8 cm (mean 4.1 cm); on the quiet simulation within 0.9 cm of perfect corrections (Questions 8.4, 9.1 and 9.2). May not: the best station on every day; WTZZ00DEU as a user unrelated to the network (it shares its site with two network stations); claims from the first hour.

These boundaries were applied to the status report's wording on 24 October 2026.

Source. the answers named.

How the answers were checked

On 24 October 2026, against branch `card/scb-clock-bed` and the runs named, each check below reproduced the answers it lists. Three statements gathered for an earlier version of the key failed their check and were removed.

Answers	Check	Result
2.1 to 2.6, 3.1, 3.2	a numpy script that builds the worked example's systems	every number of those answers; ranks 5, 13 (18 and 26 unknowns) and 7
3.4, 3.5	the parameter table of `runs/_rf0hv15_sim6`; `rf_null_space_read.py`	280 shared columns; eigenvalues 4.036e-11 and −1.803e-10, weights 0.2040 and 0.1767; cutoff 1.0459e-6
4.2, 6.3	`make_golden_config.py --verify`; the loader lines	"golden verify: PASS"; the refusal and warning as quoted
4.4	B records of G05 against the navigation file	−3.2108400641 m $= cT_{GD}$; ratio 1.646944
5.3	the datum-step script	31 satellites, 1.6278 m, 0.0027 m after the mean
5.4	`simnet_amb_integer.py --ref-free`	0.304 and 0.122 cycle
6.4, 8.3	`simnet_clock_truth_grade.py` on the card SCB cases and the baseline	every number of those answers
7.5	`simnet_wire_integer_consistency.py`; `simnet_producer_ar_grade.py`	0.252 to 0.260 against 0.250; 5572 of 5572
9.1 to 9.3	`rapid_arm.py score`; `gcensus.py`; the five-day script; `clock_stats.csv`	the hourly medians; −0.0198 m; 0.038, 0.0406, 0.018 m; 1.20, 2.23 cm
all derive answers	one script with their arithmetic	as given
removed	"the 16 further stations have no co-located network station"	KOKB00USA and network station KOKV00USA share a RINEX header position
removed	"integer fixing equals the float within 0.1 mm at ALGO00CAN"	the cited line gives another comparison
removed	"code-bias records agree to 3e-11 m"	from a scratch script; replaced by the run's own count

References

[1] P. J. de Jonge. "A processing strategy for the application of the GPS in networks". PhD thesis. Delft, The Netherlands: Delft University of Technology, 1998.

[2] D. Odijk et al. “On the estimability of parameters in undifferenced, uncombined GNSS network and PPP-RTK user models by means of S-system theory”. In: *Journal of Geodesy* 90.1 (2016), pp. 15–44.

[3] P. J. G. Teunissen. “Generalized inverses, adjustment, the datum problem and S-transformations”. In: *Optimization and Design of Geodetic Networks* (1985), pp. 11–55.